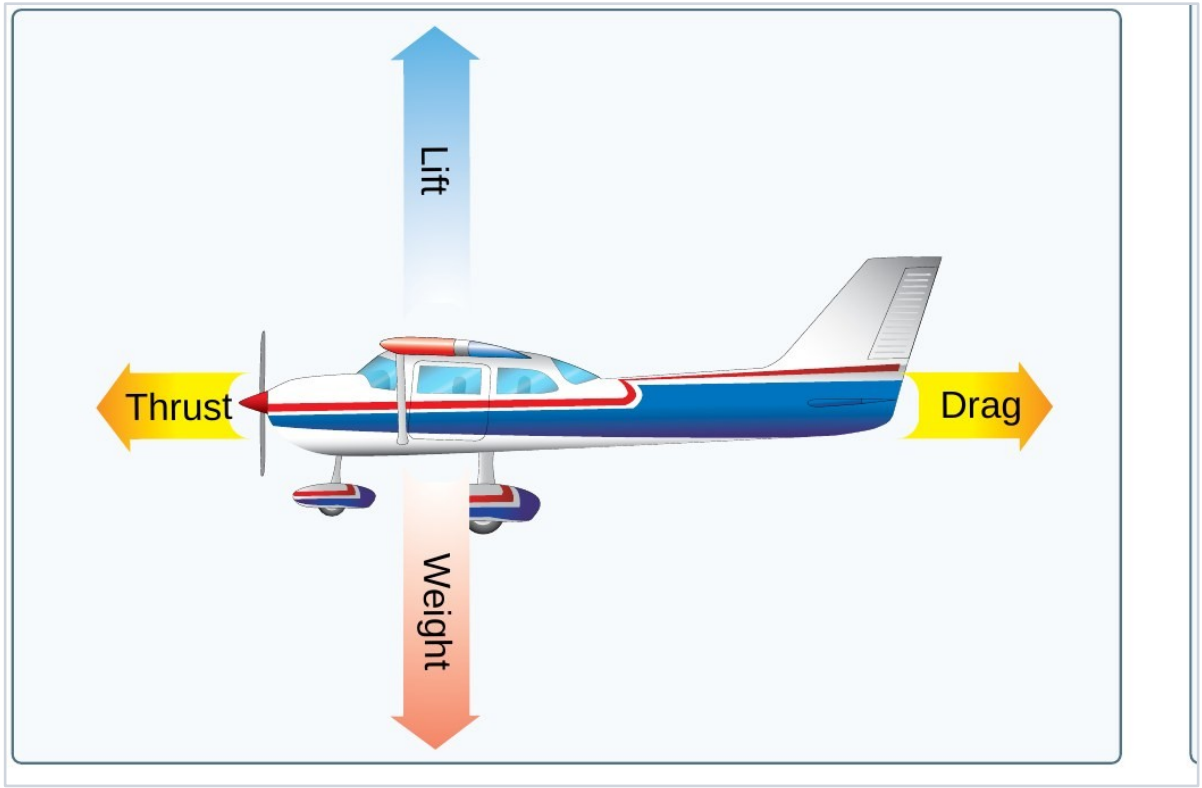
Principles of Flight

Area II: Technical Subject Areas

ACS Task II.D (AI.II.D)

Cessna 172S NAV III (G1000)

[Name], CFI Applicant

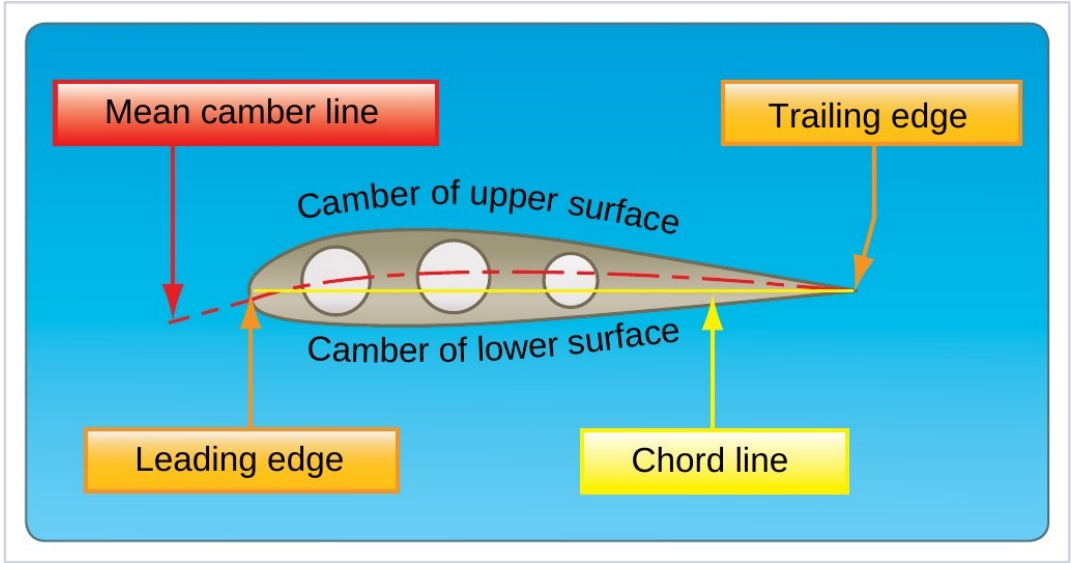


Source: PHAK Fig 5-1

Objective

- **Explain how a wing makes lift**
- **Name the four forces and drag types**
- **Explain stability and turning tendencies**
- **Know load factor and wake turbulence**

State the objective: the student will understand airfoil design, the four forces, stability, turning tendencies, load factors, and wingtip vortices, and will know how to manage the risks that come from them.



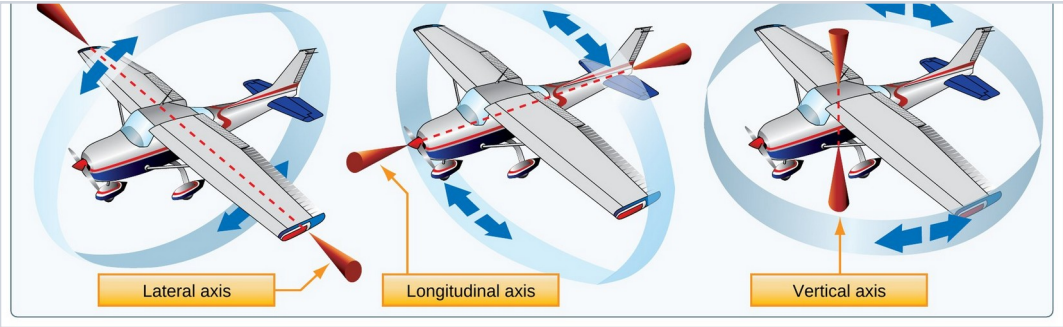
Lesson Roadmap: Each Topic Builds the Next

AI.II.D

Step	Topic	Leads to
1	Airfoil and lift (K1)	What forces act?
2	Four forces and drag (K4)	Induced drag makes vortices
3	Wingtip vortices (K6)	Why it stays stable
4	Stability and control (K2)	What pulls it off course?
5	Turning tendencies (K3)	How hard can you pull?
6	Load factors (K5)	Risk management (R1)

Show the roadmap. The order is chosen so each topic sets up the next: the wing makes lift; lift comes with forces and drag; induced drag makes wingtip vortices; then we ask why the airplane stays stable, what pulls it off course (turning tendencies), and how much load it can take. Risk management ties it all together.

Teaching the order this way follows the AIH idea of building from the known to the unknown and connecting each new idea to the last one.

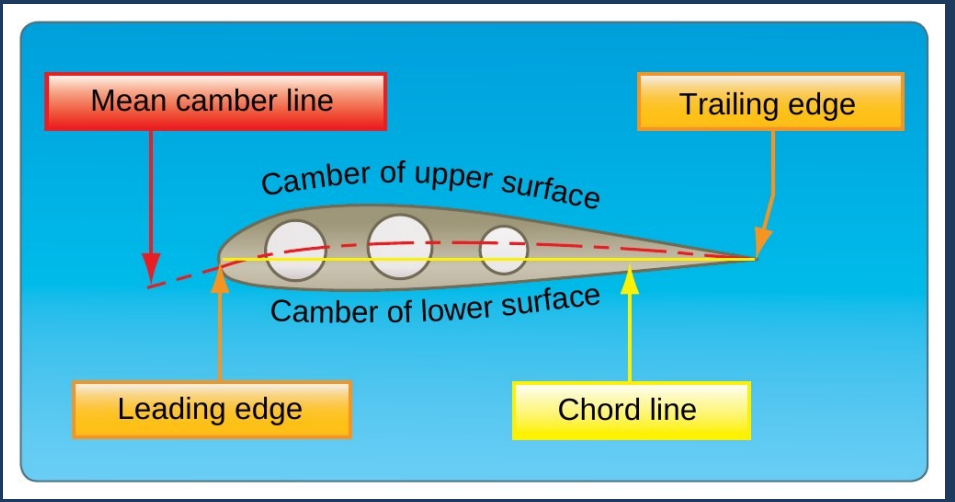


NEXT UP

Start with step 1: how the wing makes lift.

Step 1 of 6 | AI.II.D.K1

Airfoil Design and Lift



What you will learn

- Parts of an airfoil
- Bernoulli and Newton
- Relative wind and AOA
- Critical AOA and the stall

Start here: everything else in this lesson comes from how the wing makes lift.

Source: PHAK 4-6 to 4-8, PHAK 5-4

Definitions: Chord Line and Camber

AI.II.D.K1

CHORD LINE

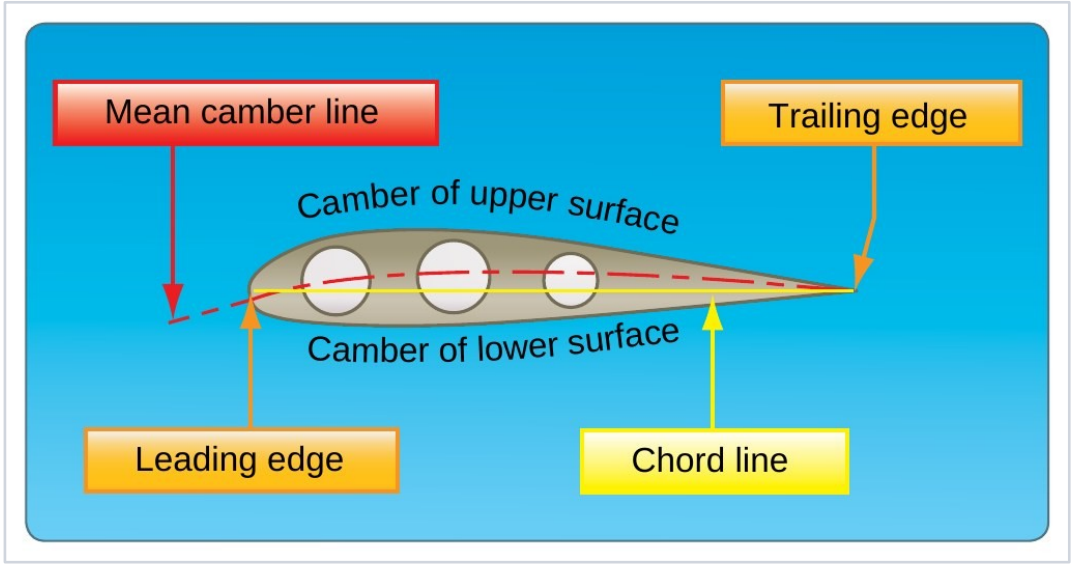
“a straight line drawn through the profile connecting the extremities of the leading and trailing edges.”

CAMBER

“The camber of an airfoil is the characteristic curve of its upper and lower surfaces. The upper camber is more pronounced, while the lower camber is comparatively flat.”

TEACHING NOTE

The glossary adds that the curved upper camber makes the airflow above the wing much faster than below it (PHAK G-6).



Source: PHAK 4-6, PHAK G-6, PHAK Fig 4-5

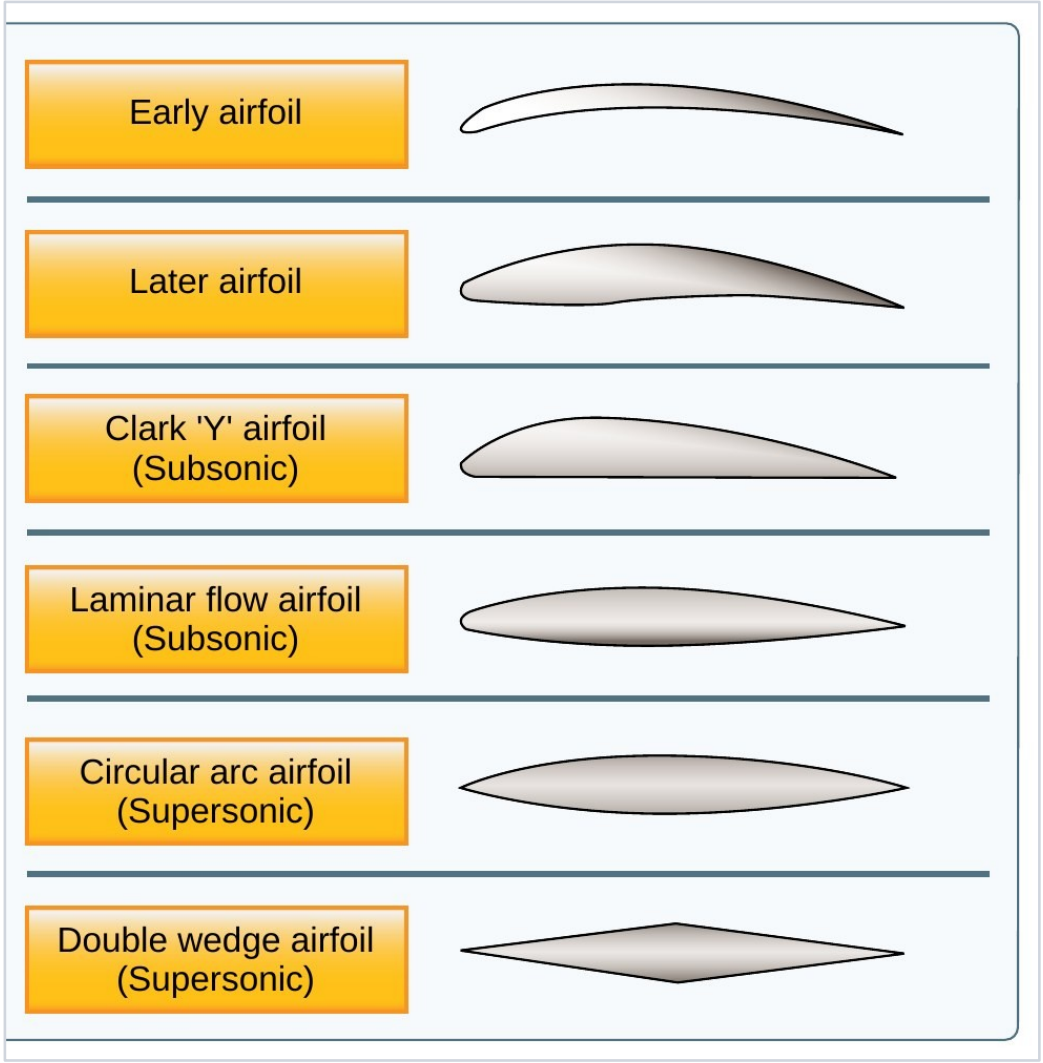
Airfoil Design Characteristics

AI.II.D.K1

- **Shape is built to make lift**
- **Thick, curved:** good at low speed
- **Thin:** better for high speed
- **Every design is a tradeoff**

An airfoil is shaped to take advantage of the air's response to physical laws, giving a positive pressure lifting action from below and a negative pressure lifting action from above (PHAK 4-7).

Figure 4-6 shows the progression from early airfoils to laminar flow and supersonic shapes. Each design is chosen for the airplane's purpose.



Source: PHAK 4-6, PHAK 4-7, PHAK Fig 4-6

Definition: Bernoulli's Principle

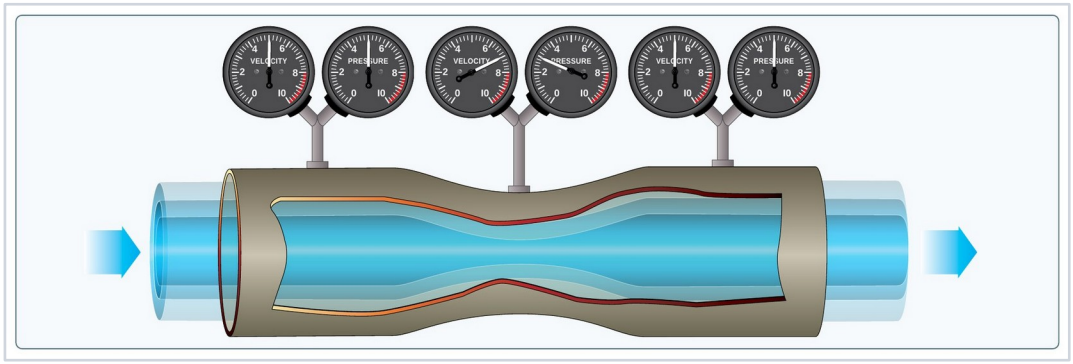
AI.II.D.K1

BERNOULLI'S PRINCIPLE

“as the velocity of a moving fluid (liquid or gas) increases, the pressure within the fluid decreases.”

IN PLAIN WORDS

Air speeds up over the curved top of the wing, so the pressure there drops. Higher pressure underneath pushes the wing up.



MEMORY AID

Speed up, pressure down.

Source: PHAK 4-6, PHAK Fig 4-4

Newton's Third Law and Downwash

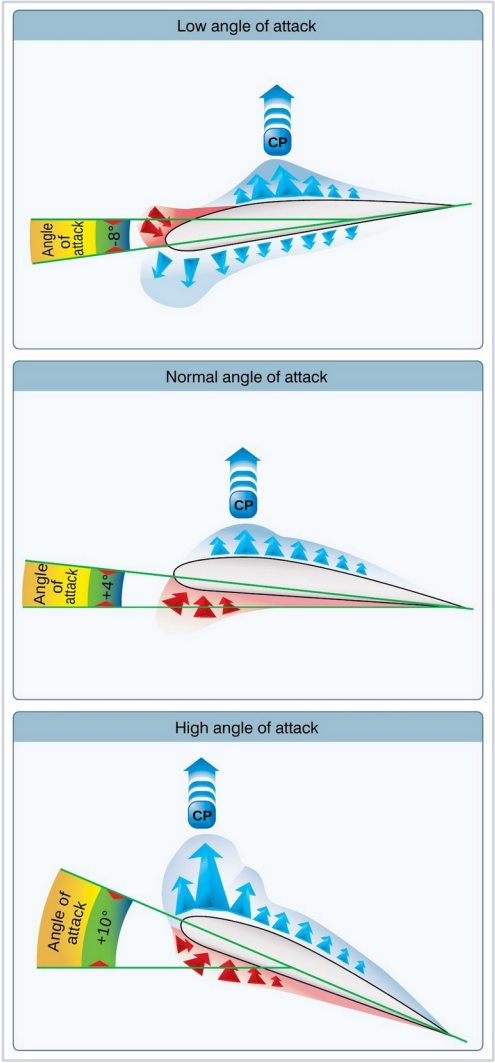
AI.II.D.K1

- **Equal and opposite reaction**
- **Wing pushes air down**
- **Air pushes wing up**
- **Both laws work together**

Newton's Third Law: "For every action, there is an equal and opposite reaction" (PHAK 4-3).

The downward backward flow from the top surface creates a downwash. Applying Newton's third law, the reaction to this downward backward flow is an upward forward force on the airfoil (PHAK 4-8).

Both Bernoulli's Principle and Newton's Laws are in operation whenever lift is being generated (PHAK 4-8). Figure 4-7 shows the pressure pattern changing as AOA changes.



Source: PHAK 4-3, PHAK 4-8, PHAK Fig 4-7

Definitions: Relative Wind and Angle of Attack

AI.II.D.K1

RELATIVE WIND

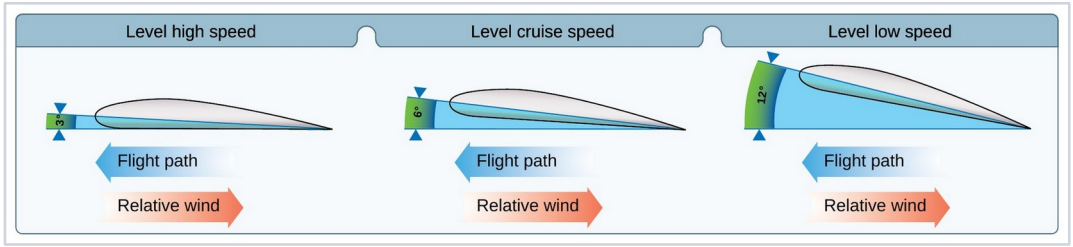
“Direction of the airflow produced by an object moving through the air. The relative wind for an airplane in flight flows in a direction parallel with and opposite to the direction of flight.”

ANGLE OF ATTACK

“the angle at which relative wind meets an airfoil. It is the angle that is formed by the chord of the airfoil and the direction of the relative wind or between the chord line and the flight path.”

TEACHING NOTE

Lift is perpendicular to the relative wind (PHAK 5-8), which your whiteboard notes also show.



KNOW THIS FOR THE CHECKRIDE

Lift always acts perpendicular to the relative wind, and the relative wind comes from the flight path, not the nose.

Source: PHAK G-25, PHAK G-3, PHAK Fig 5-3

Lift, Speed, and AOA

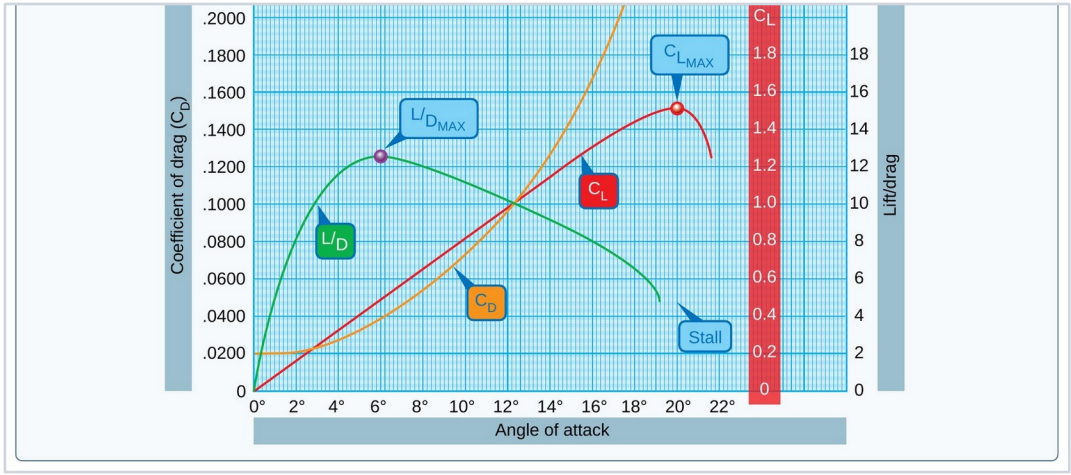
AI.II.D.K1

- **Lift:** proportional to speed squared
- **Double speed:** four times lift
- **Slower:** need higher AOA
- **Pilot controls lift with AOA**

Lift is proportional to the square of the velocity. An airplane at 200 knots has four times the lift of the same airplane at 100 knots, if AOA and other factors stay the same (PHAK 5-4).

The lift equation is L = CL x density x V squared x S, divided by 2 (PHAK 5-4). The pilot controls lift mainly by changing AOA and airspeed.

Figure 5-5 shows the coefficient of lift rising with AOA until it peaks, then falling off: that is the stall.



KNOW THIS FOR THE CHECKRIDE

Lift is proportional to the square of the velocity. Twice the speed, four times the lift (PHAK 5-4).

Source: PHAK 5-4, PHAK Fig 5-5

Definition: Critical Angle of Attack

AI.II.D.K1

CRITICAL ANGLE OF ATTACK

“The angle of attack at which a wing stalls regardless of airspeed, flight attitude, or weight.”

IN PLAIN WORDS

A wing stalls because the AOA is too high, not because the airspeed is too low. Stall speed changes with weight, bank, and flaps. The critical AOA does not.

STALL SPEEDS AT 2550 POUNDS

CONDITIONS:
Power Off

MOST REARWARD CENTER OF GRAVITY

FLAP SETTING	ANGLE OF BANK							
	0°		30°		45°		60°	
	KIAS	KCAS	KIAS	KCAS	KIAS	KCAS	KIAS	KCAS
UP	48	53	52	57	57	63	68	75
10°	42	50	45	54	50	59	59	71
FULL	40	48	43	52	48	57	57	68

MOST FORWARD CENTER OF GRAVITY

FLAP SETTING	ANGLE OF BANK							
	0°		30°		45°		60°	
	KIAS	KCAS	KIAS	KCAS	KIAS	KCAS	KIAS	KCAS
UP	48	53	52	57	57	63	68	75
10°	43	51	46	55	51	61	61	72
FULL	40	48	43	52	48	57	57	68

NOTE

- Altitude loss during a stall recovery may be as much as 230 feet.
- KIAS values are approximate.

Source: PHAK G-9, POH Fig 5-3

Quick Check

AI.II.D.K1

1 **State Bernoulli's Principle.**

Answer: As the velocity of a moving fluid increases, the pressure within the fluid decreases (PHAK 4-6).

2 **If you double your airspeed at the same AOA, what happens to lift?**

Answer: It becomes four times as much (PHAK 5-4).

3 **Can a wing stall at high airspeed?**

Answer: Yes. It stalls at the critical AOA regardless of airspeed, attitude, or weight (PHAK G-9).

NEXT UP

Next up, the four forces. The wing makes lift, but lift never comes alone. What else is acting on the airplane?

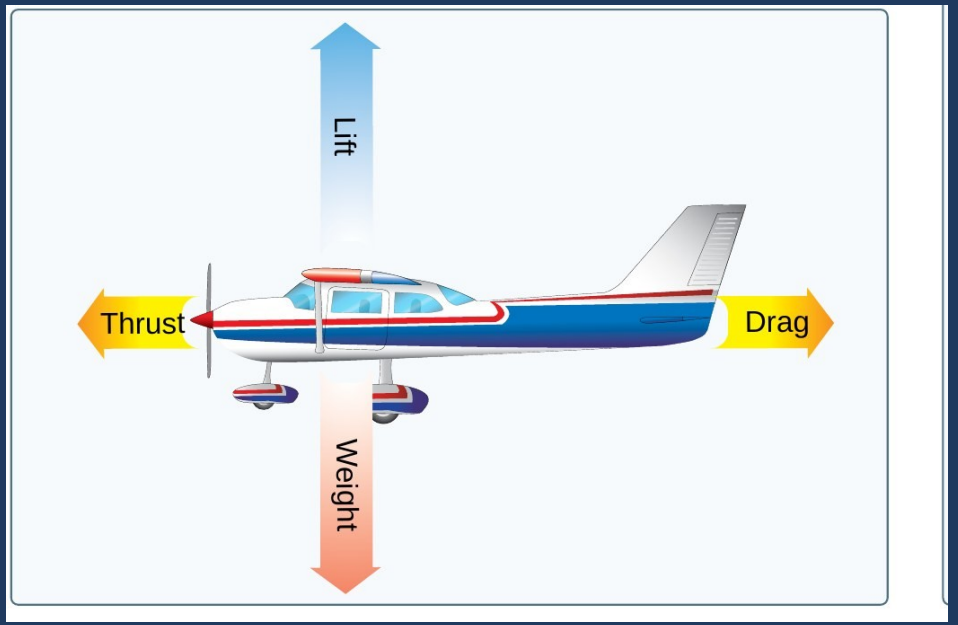
Source: PHAK 4-6, PHAK 4-8, PHAK 5-4, PHAK G-9

Step 2 of 6 | AI.II.D.K4

Forces Acting on an Airplane

What you will learn

- Thrust, drag, lift, weight
- Forces in balance and in a climb
- Parasite drag: form, interference, skin friction
- Induced drag and L/D max



Coming from step 1: the wing makes lift, but lift never comes alone. Here are the forces that come with it.

Source: PHAK 5-1 to 5-8

The Four Forces

AI.II.D.K4

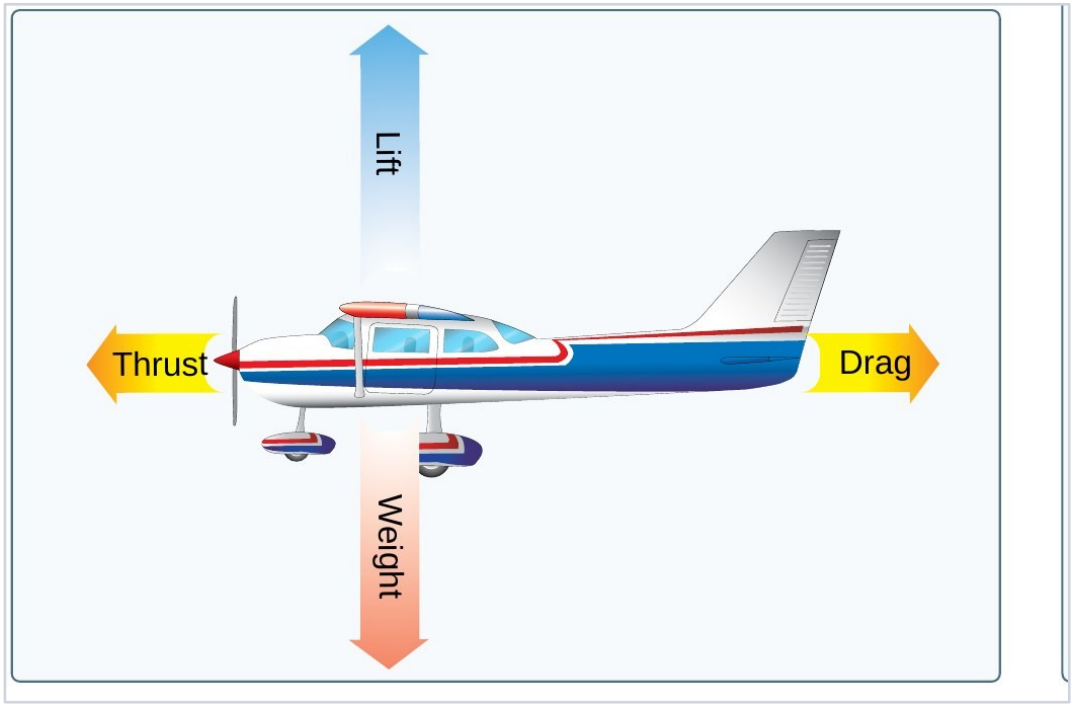
- **Thrust:** forward, from the prop
- **Drag:** rearward, from airflow
- **Lift:** perpendicular to flight path
- **Weight:** down through the CG

Thrust: the forward force produced by the powerplant/propeller. It opposes or overcomes drag (PHAK 5-1).

Drag: a rearward, retarding force caused by disruption of airflow by the wing, fuselage, and other protruding objects (PHAK 5-1).

Lift: produced by the dynamic effect of the air acting on the airfoil, acting perpendicular to the flight path through the center of lift (PHAK 5-1).

Weight: the combined load of the aircraft, crew, fuel, and cargo, acting vertically downward through the CG (PHAK 5-2).



Source: PHAK 5-1, PHAK 5-2, PHAK Fig 5-1

Forces in Balance and in a Climb

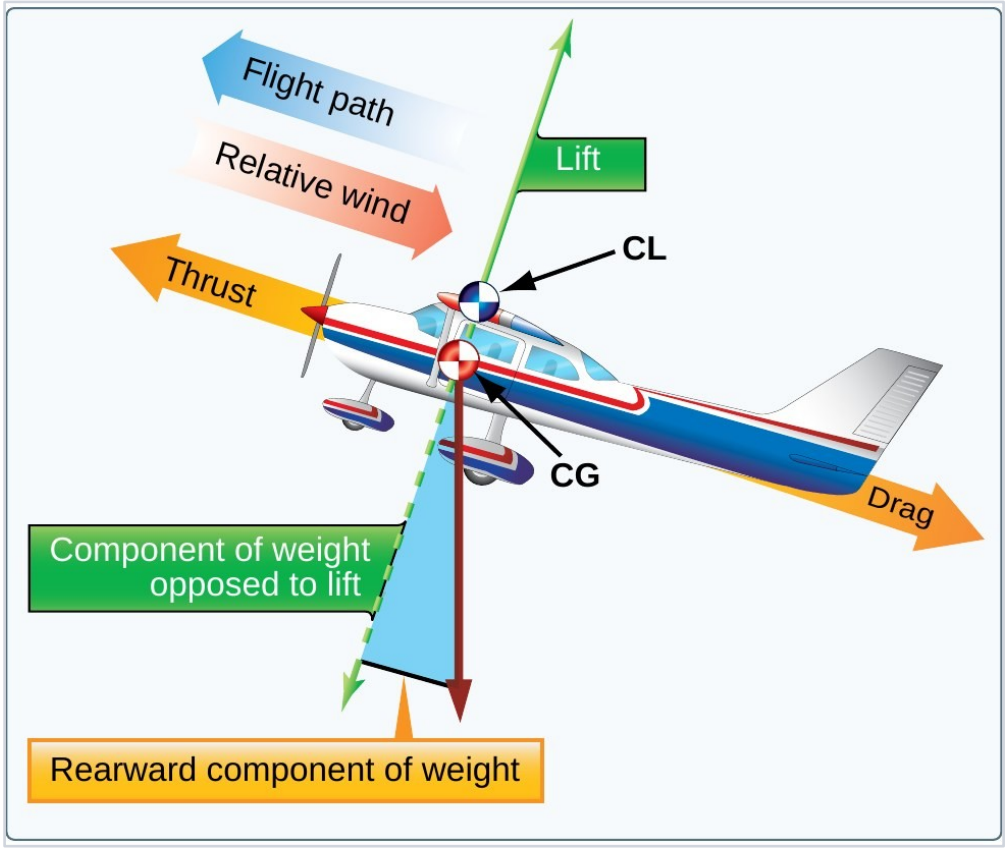
AI.II.D.K4

- **Steady flight:** forces cancel
- **Not all four are equal**
- **Lift and weight are larger**
- **Climb:** weight acts partly as drag

In steady flight, the sum of opposing forces is always zero. That does not mean the four forces are equal; it means the opposing forces cancel (PHAK 5-2).

In straight, level, unaccelerated flight, lift and weight are equal and are greater than thrust and drag, which are equal only to each other (PHAK 5-2).

Figure 5-2: in a climb, part of the weight acts rearward like drag, so more thrust is needed.



KNOW THIS FOR THE CHECKRIDE

Opposing forces cancel. Lift and weight are bigger than thrust and drag (PHAK 5-2).

Source: PHAK 5-2, PHAK Fig 5-2

Definition: Parasite Drag

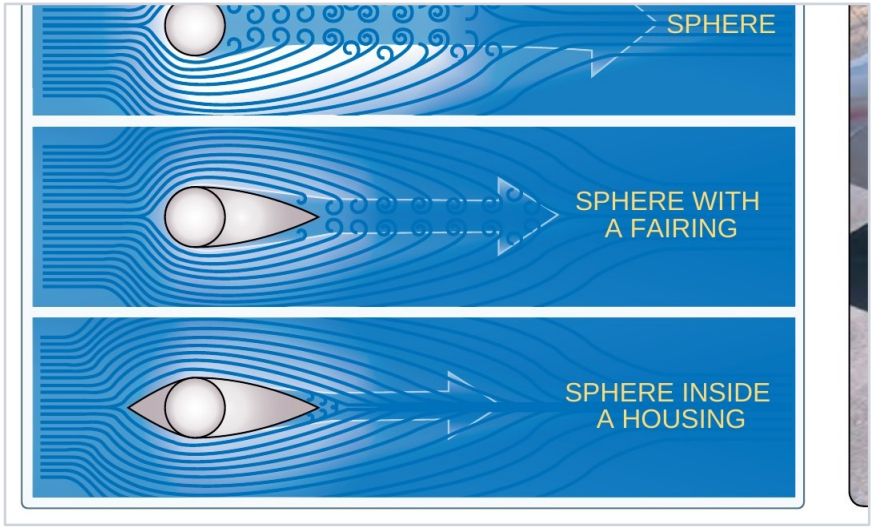
AI.II.D.K4

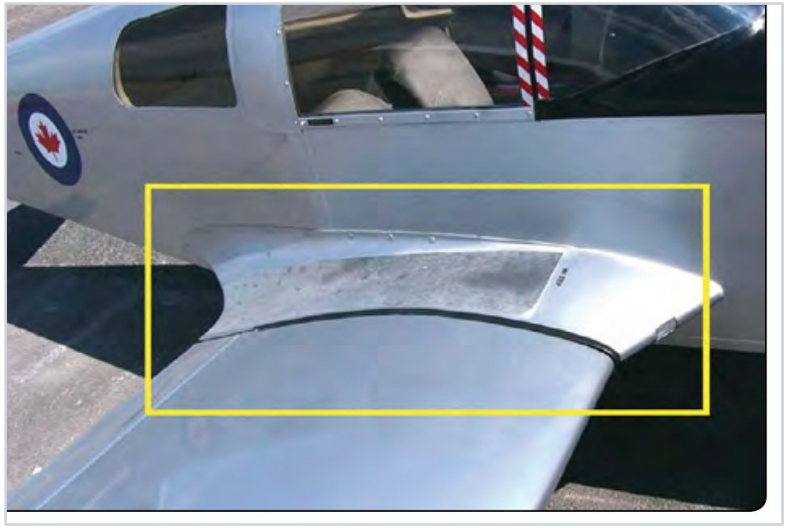
PARASITE DRAG

“Parasite drag is comprised of all the forces that work to slow an aircraft's movement. As the term parasite implies, it is the drag that is not associated with the production of lift.”

IN PLAIN WORDS

Three types: form drag (shape), interference drag (airstreams meeting, like wing and fuselage), and skin friction (rough surfaces, rivets, dirt).





Source: PHAK 5-6, PHAK Fig 5-7, PHAK Fig 5-8

Induced Drag and the Drag Curve

AI.II.D.K4

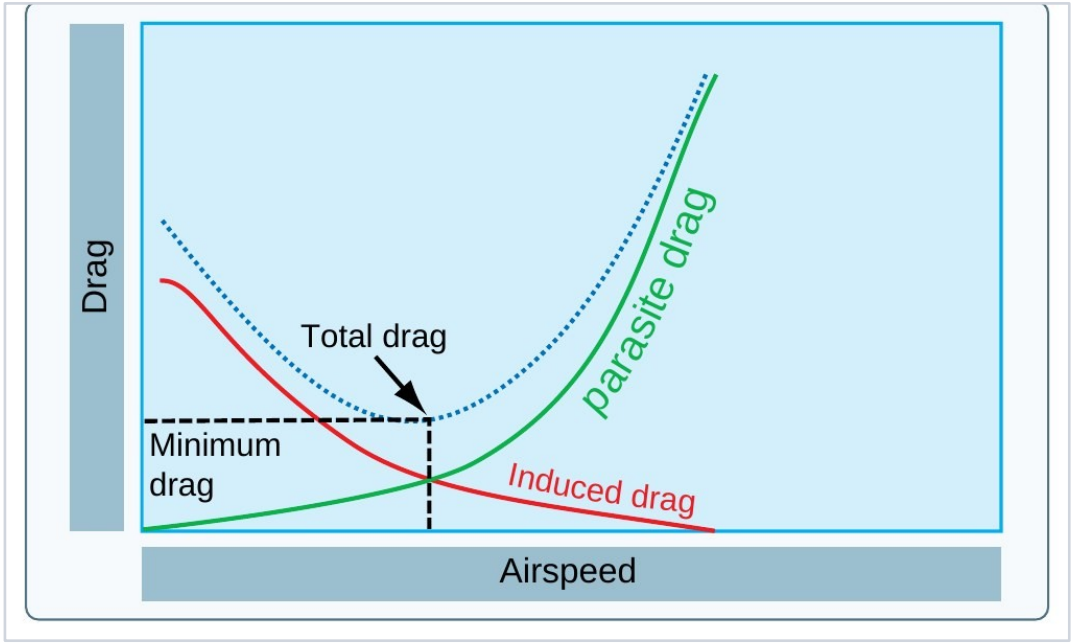
- **Induced:** comes from making lift
- **More AOA, more induced drag**
- **Parasite rises with speed squared**
- **L/D max:** lowest total drag

Induced drag is inherent whenever an airfoil is producing lift and is inseparable from the production of lift (PHAK 5-7).

As AOA increases, induced drag increases. The lower the airspeed, the greater the AOA needed, so the greater the induced drag. Induced drag varies inversely with the square of the airspeed; parasite drag increases as the square of the airspeed (PHAK 5-8).

At L/D max, total drag is at its minimum (PHAK 5-5).

Note on your whiteboard: the PHAK says induced drag varies inversely with the square of airspeed and parasite drag increases as the square of airspeed, not just "inversely" and "directly."



KNOW THIS FOR THE CHECKRIDE

Induced drag: inversely with the square of airspeed.
Parasite drag: as the square of airspeed (PHAK 5-8).

Source: PHAK 5-7, PHAK 5-8, PHAK 5-5, PHAK Fig 5-6

Quick Check

AI.II.D.K4

1 **In straight and level flight, are all four forces equal?**

Answer: No. Lift equals weight and thrust equals drag, but lift and weight are larger (PHAK 5-2).

2 **Name the three types of parasite drag.**

Answer: Form, interference, and skin friction (PHAK 5-6).

3 **You slow down toward a stall. Which drag rises sharply?**

Answer: Induced drag (PHAK 5-8).

NEXT UP

Next up, wingtip vortices. Induced drag comes from air spilling around the wingtip. Where does that air go?

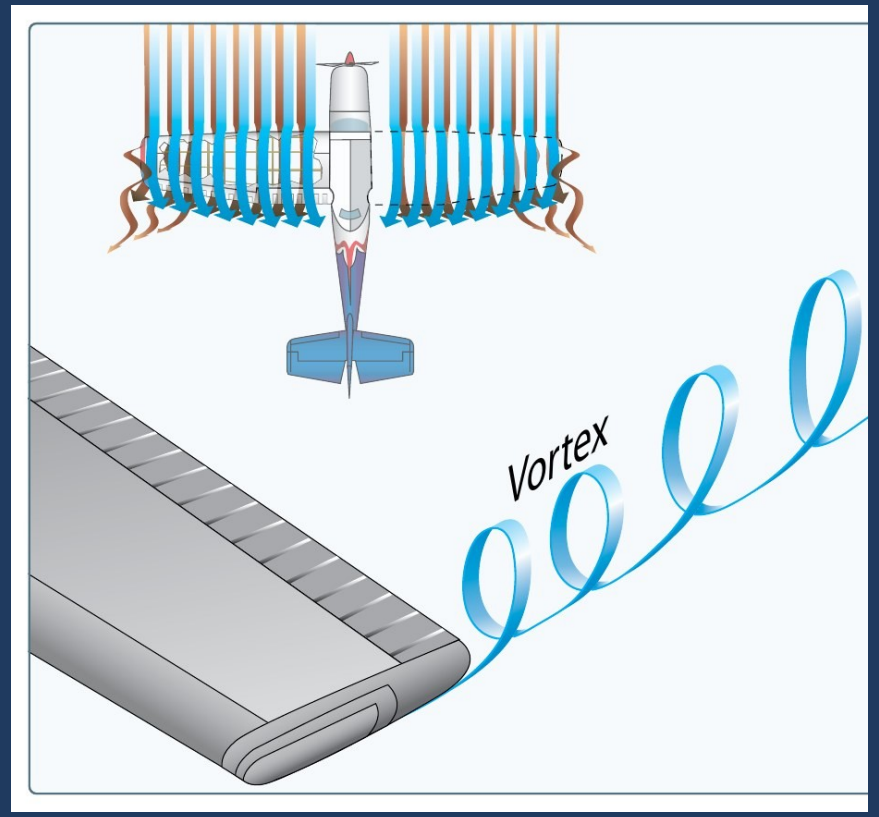
Source: PHAK 5-2, PHAK 5-6, PHAK 5-8

Step 3 of 6 | AI.II.D.K6

Wingtip Vortices

What you will learn

- How vortices form
- How they move and sink
- Wake turbulence precautions



Coming from step 2: induced drag comes from air spilling around the wingtip. That spill is a wingtip vortex.

Source: PHAK 5-7 to 5-10, AIM 7-4

How Wingtip Vortices Form

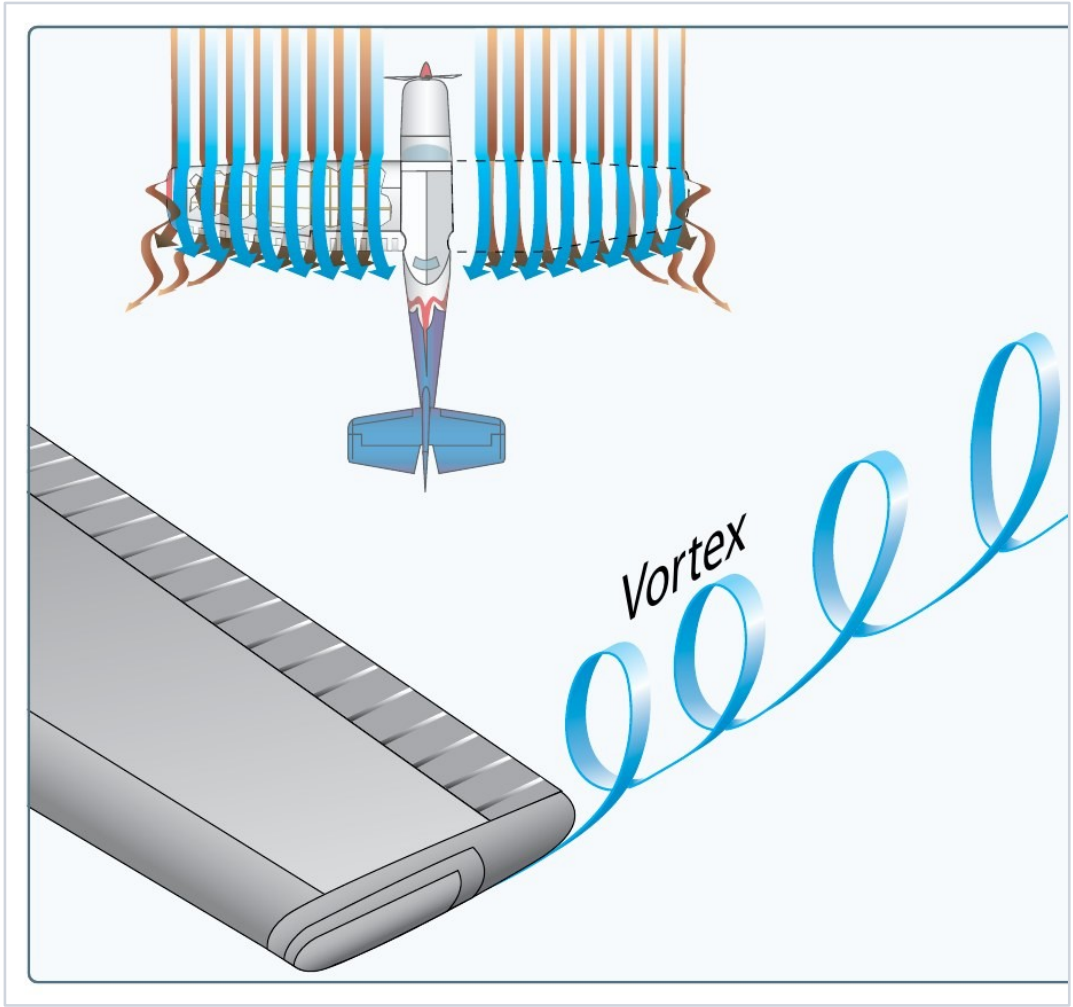
AI.II.D.K6

- **High pressure under the wing**
- **Air spills around the tip**
- **Forms a spinning vortex**
- **Strongest:** heavy, clean, slow

Below the wing the pressure is higher than above it. The air flows from the high pressure area below the tip up to the low pressure area on top, creating vortices that trail behind the airfoil (PHAK 5-7).

As the air curls upward around the tip, it combines with the downwash to form a fast-spinning trailing vortex (PHAK 5-9). Your whiteboard note matches this wording.

Vortex strength is directly proportional to weight and inversely proportional to wingspan and speed. Strongest during takeoff, climb, and landing (PHAK 5-9).



Source: PHAK 5-7, PHAK 5-8, PHAK 5-9, PHAK Fig 5-12

How Vortices Move

AI.II.D.K6

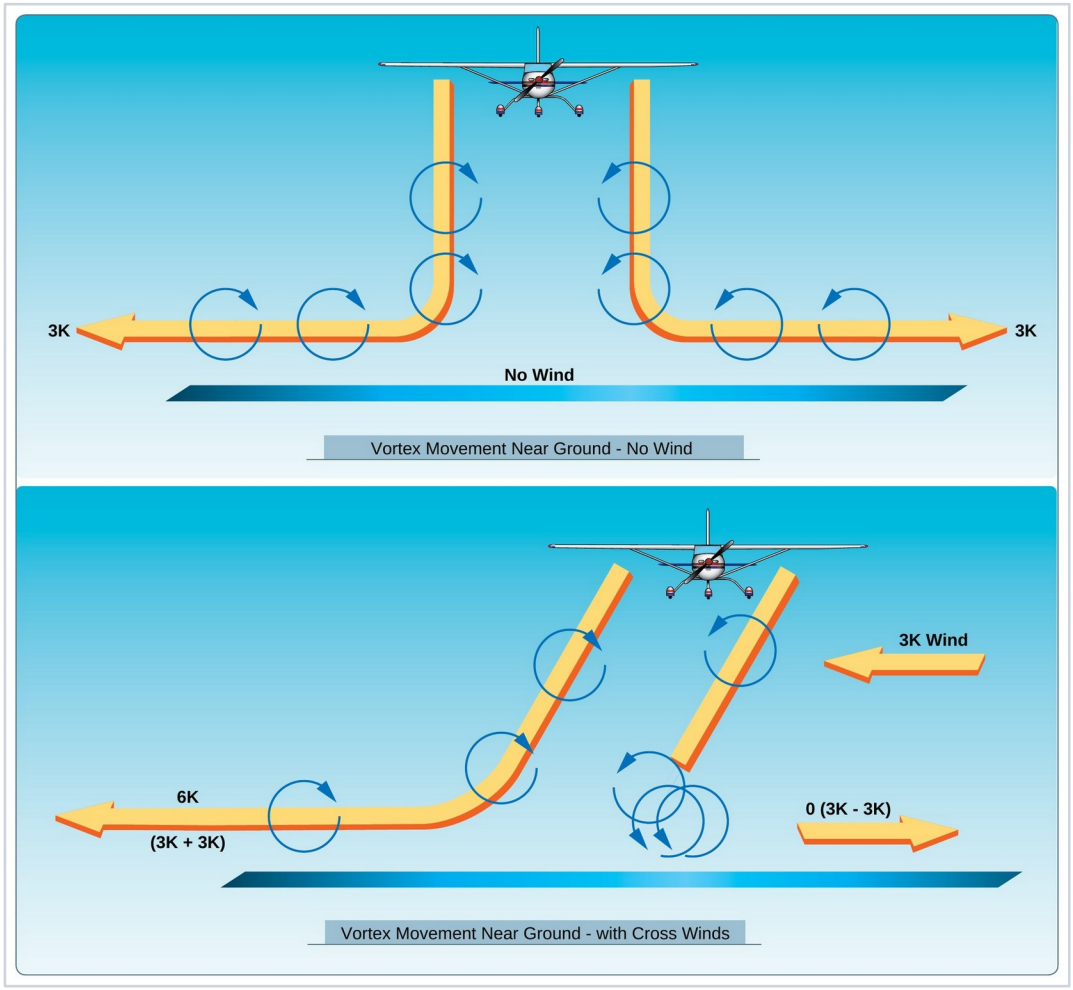
- **From rotation to touchdown**
- **Sink several hundred fpm**
- **Near ground:** drift 2 to 3 knots
- **Light crosswind can hold one**

An aircraft makes vortices from the moment it rotates on takeoff until touchdown, since they are a by-product of lift (AIM 7-4-4).

Vortices from larger aircraft sink several hundred feet per minute, slowing and weakening with time and distance (AIM 7-4-4).

Within 100 to 200 feet of the ground, they move laterally at 2 or 3 knots. A light crosswind of 1 to 5 knots can hold the upwind vortex in the touchdown zone (PHAK Fig 5-15).

Vortices drift with the wind. A 10 knot wind moves them about 1,000 feet in a minute (PHAK 5-9).



KNOW THIS FOR THE CHECKRIDE

A light quartering crosswind is the worst case. It can hold a vortex over the runway.

Source: AIM 7-4-4, PHAK 5-9, PHAK Fig 5-15

Wake Turbulence Precautions

AI.II.D.K6

- **Landing:** stay above, land beyond
- **Departing:** rotate before them
- **Climb above their path**
- **En route:** avoid below and behind

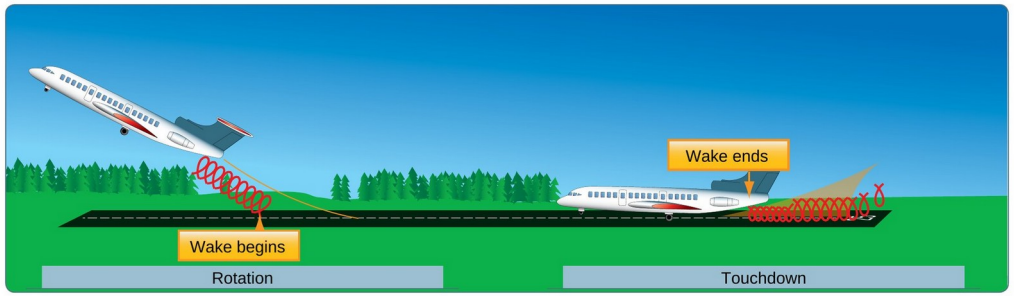
Landing behind a larger aircraft on the same runway: stay at or above its final approach path, note its touchdown point, and land beyond it (AIM 7-4-6).

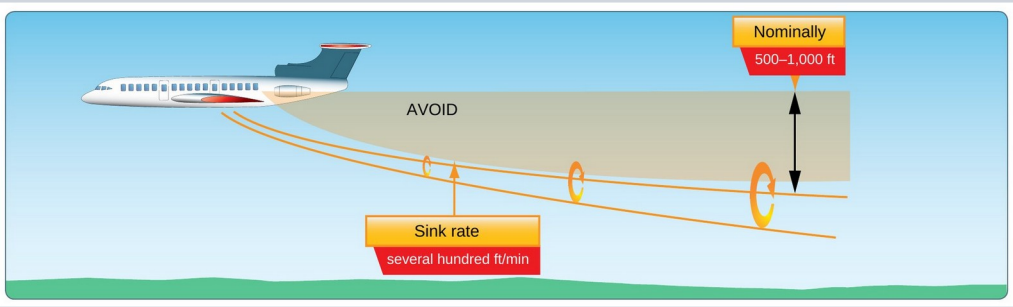
Departing behind a larger aircraft: note its rotation point, rotate prior to it, and climb above its climb path until turning clear (AIM 7-4-6).

After a larger aircraft does a low approach, missed approach, or touch-and-go, wait at least 2 minutes before your takeoff or landing (AIM 7-4-6).

En route: avoid flight below and behind a large aircraft's path. Avoid following within 1,000 feet below (AIM 7-4-6, PHAK 5-9).

Whether or not ATC gives a wake turbulence caution, the pilot is expected to adjust to avoid wake encounters (AIM 7-4-6).





MEMORY AID

Land beyond, lift off before, stay above.

REAL WORLD SCENARIO

You are cleared for takeoff right after a regional jet departs. Ask your student where they plan to rotate and which way to turn after takeoff.

Source: AIM 7-4-6, PHAK 5-9

Quick Check

AI.II.D.K6

1 **When are wingtip vortices strongest?**

Answer: When the generating aircraft is heavy, clean, and slow (PHAK 5-9).

2 **Landing behind a larger aircraft on the same runway: where do you touch down?**

Answer: Beyond its touchdown point, staying at or above its approach path (AIM 7-4-6).

3 **A larger aircraft just did a touch-and-go. How long do you wait?**

Answer: At least 2 minutes (AIM 7-4-6).

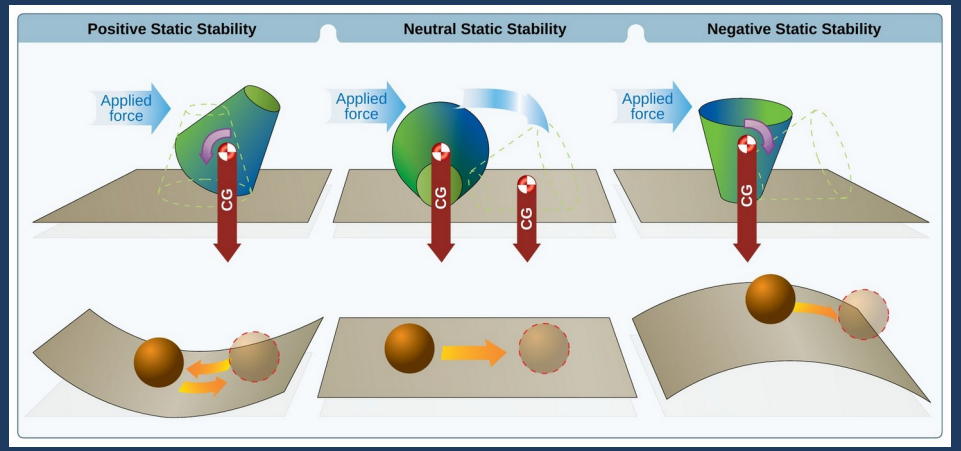
NEXT UP

Next up, stability. We know the forces. Why does the airplane stay balanced when a gust hits it?

Source: PHAK 5-9, AIM 7-4-4, AIM 7-4-6

Step 4 of 6 | AI.II.D.K2

Stability, Maneuverability, Controllability



What you will learn

- Static and dynamic stability
- Pitch, roll, and yaw stability
- Maneuverability and controllability
- How CG changes them

Coming from step 3: we know what forces act. Now, why does the airplane stay in balance when a gust hits it?

Source: PHAK 5-14 to 5-19

Definition: Stability

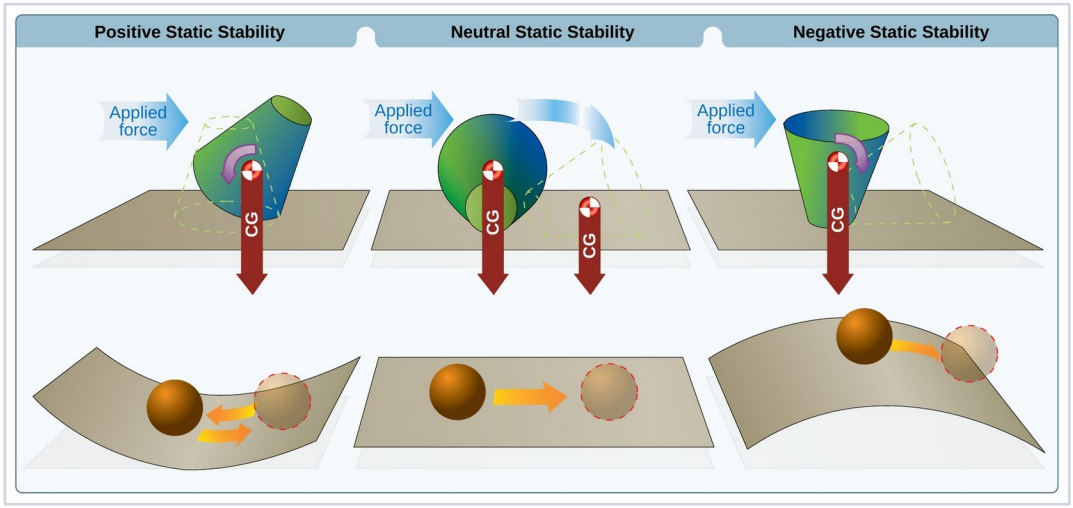
AI.II.D.K2

STABILITY

“Stability is the inherent quality of an aircraft to correct for conditions that may disturb its equilibrium and to return to or to continue on the original flight path. It is primarily an aircraft design characteristic.”

IN PLAIN WORDS

Static stability is the first reaction (ball in a bowl rolls back). Dynamic stability is what happens over time (the bounces get smaller).



MEMORY AID

Bowl, table, hill: positive, neutral, negative.

Source: PHAK 5-14, PHAK Fig 5-21

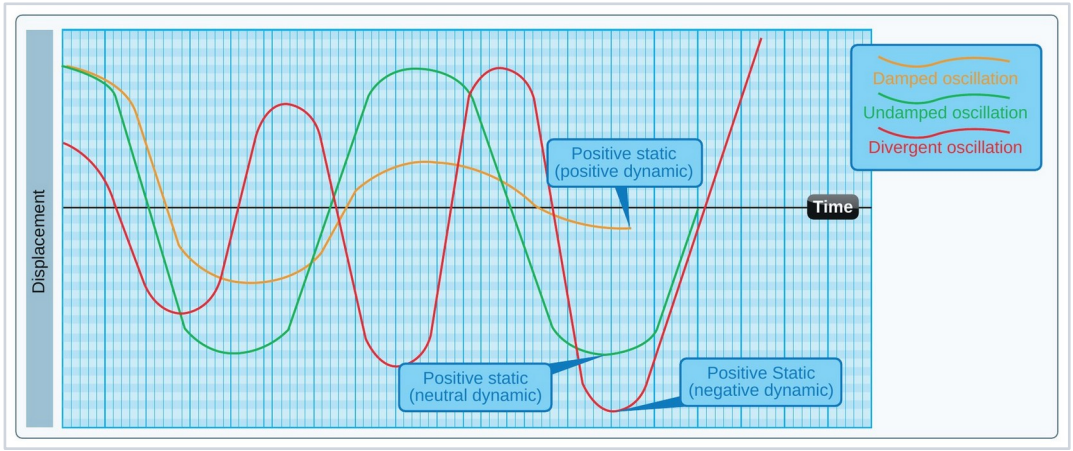
Dynamic Stability

AI.II.D.K2

- **Response over time**
- **Positive:** swings get smaller
- **Neutral:** swings stay the same
- **Negative:** swings grow

Dynamic stability is the aircraft response over time. Positive: motion decreases. Neutral: neither decreases nor increases (like a worn shock absorber). Negative: motion increases (PHAK 5-15).

In flight demonstration: trim for hands-off level flight and give the controls a slight push. If the oscillations gradually stop, the airplane has positive stability; if they continue, neutral; if they increase, unstable (PHAK 5-17).



Source: PHAK 5-14, PHAK 5-15, PHAK 5-17, PHAK Fig 5-22

Longitudinal Stability (Pitch)

AI.II.D.K2

- **About the lateral axis**
- **CG ahead of center of lift**
- **Tail pushes down to balance**
- **Slow down:** nose drops

Longitudinal stability is stability about the lateral axis: pitch (PHAK 5-15).

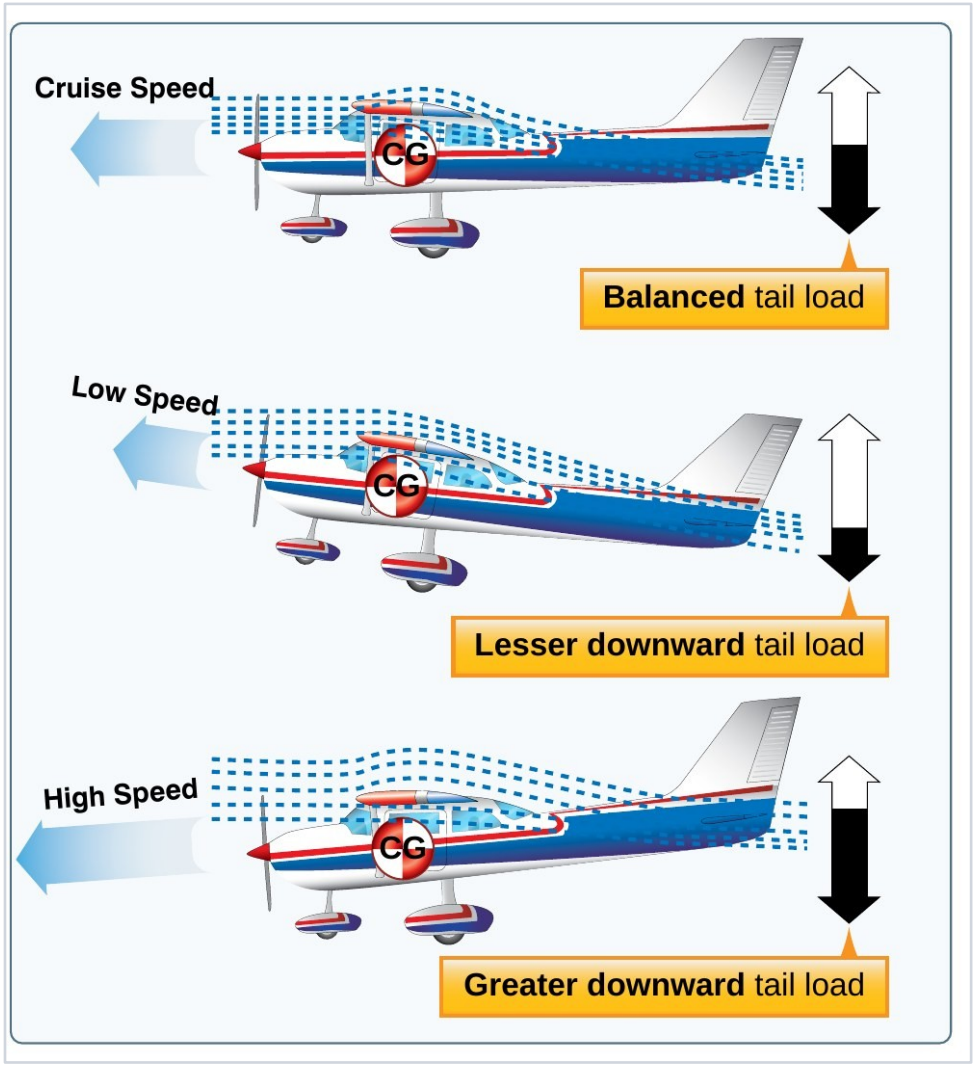
It depends on the location of the wing and the horizontal tail relative to the CG, and on the size of the tail (PHAK 5-15).

Most airplanes have the center of lift behind the CG, so they are nose heavy. The horizontal stabilizer makes a downward force to balance it (PHAK 5-16).

If speed decreases, downwash on the tail decreases, the nose pitches down, and airspeed increases again. This is why the C172S tends to return to its trimmed speed (PHAK 5-16).

KNOW THIS FOR THE CHECKRIDE

CG ahead of the center of lift plus a tail-down force: the airplane tries to return to a safe attitude (PHAK 5-17).



Source: PHAK 5-15, PHAK 5-16, PHAK Fig 5-24

Lateral Stability (Roll)

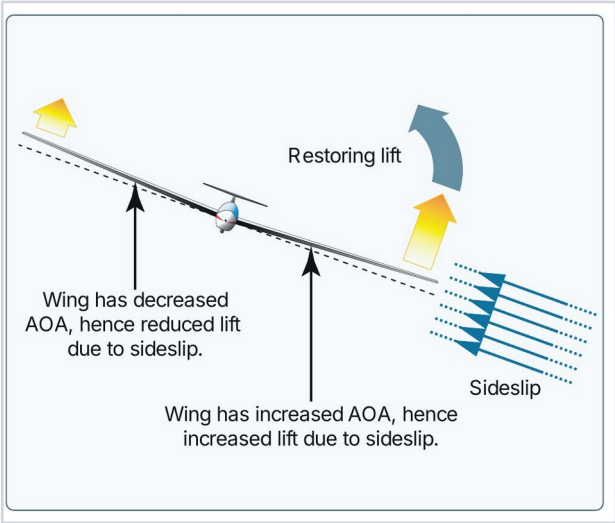
AI.II.D.K2

- **About the longitudinal axis**
- **Dihedral, sweepback, keel effect**
- **Low wing gets more AOA**
- **High wing helps (C172S)**

Lateral stability is stability about the longitudinal axis: roll. Four design factors: dihedral, sweepback, keel effect, and weight distribution (PHAK 5-17).

Dihedral: when a gust rolls the airplane, it sideslips. The wing slipping into the wind has a higher AOA and more lift, which rolls the airplane back (PHAK 5-18).

A high wing airplane like the C172S gets lateral stability from keel effect: the fuselage hangs below the wing like a pendulum (PHAK 5-18).





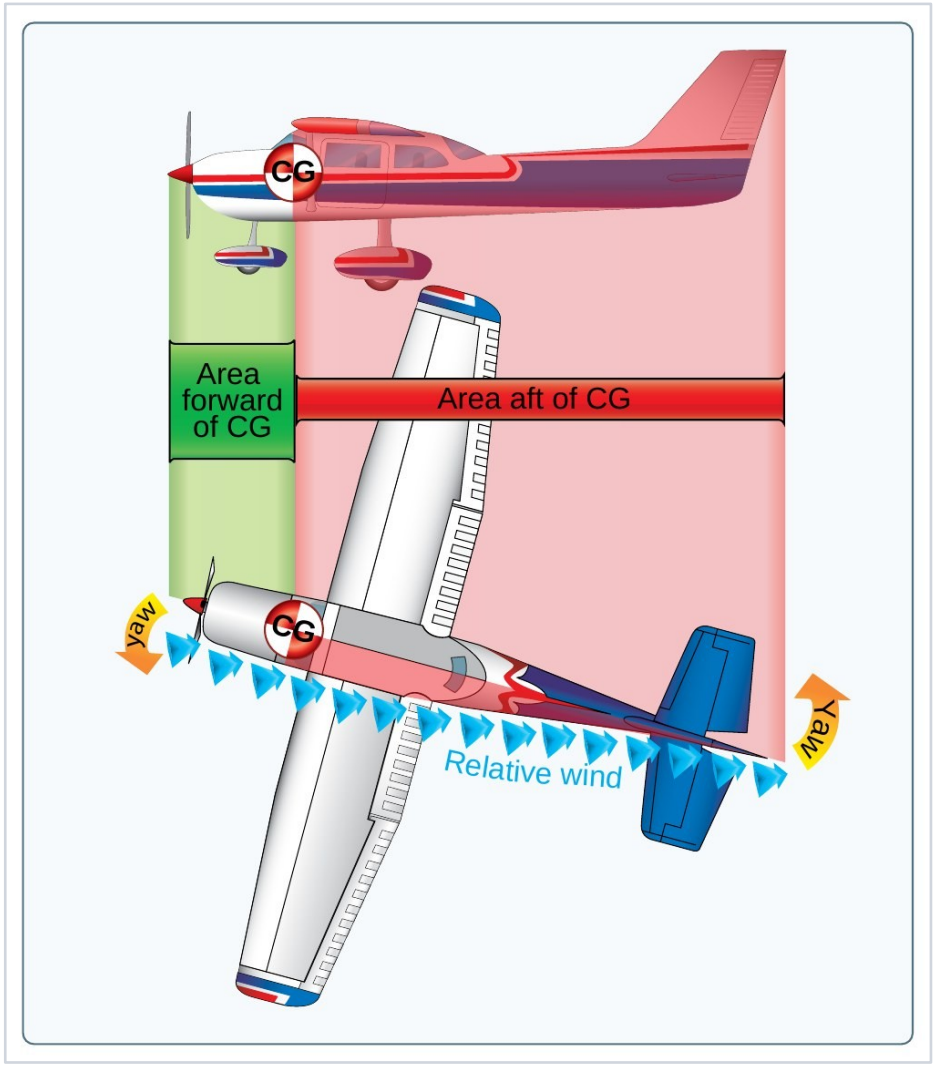
Directional Stability (Yaw)

AI.II.D.K2

- **About the vertical axis**
- **Vertical fin acts like arrow feathers**
- **More side area behind the CG**
- **Weathervanes into the wind**

Directional stability is stability about the vertical axis: yaw. It is the most easily achieved stability in design (PHAK 5-19).

The vertical fin and fuselage sides aft of the CG make the airplane act like a weather vane or arrow, pointing into the relative wind. The farther aft and larger the fin, the greater the directional stability (PHAK 5-19).



MEMORY AID

Weather vane or arrow: more area behind the pivot point.

Source: PHAK 5-19, PHAK Fig 5-32

Definitions: Maneuverability and Controllability

AI.II.D.K2

MANEUVERABILITY

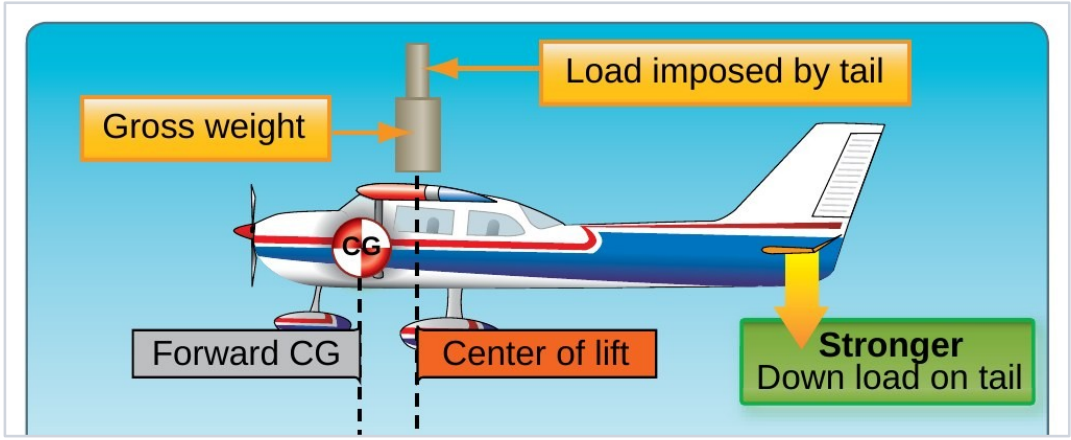
"the quality of an aircraft that permits it to be maneuvered easily and to withstand the stresses imposed by maneuvers."

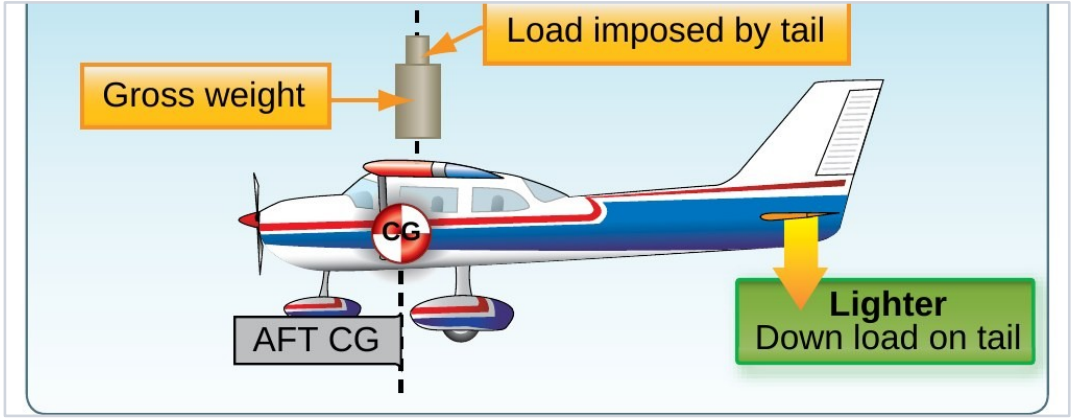
CONTROLLABILITY

"the capability of an aircraft to respond to the pilot's control, especially with regard to flight path and attitude."

IN PLAIN WORDS

Forward CG: more stable, higher stall speed, heavier controls. Aft CG: less stable, lighter controls, harder stall recovery.





Source: PHAK 5-15, PHAK 5-43, PHAK 5-44

Quick Check

AI.II.D.K2

1 **The nose returns toward level after a push, then the swings die out. Static and dynamic stability?**

Answer: Positive static and positive dynamic (PHAK 5-14, PHAK 5-17).

2 **Which axis is pitch stability about?**

Answer: The lateral axis (PHAK 5-15).

3 **What makes a C172S laterally stable?**

Answer: Dihedral and keel effect from the high wing (PHAK 5-18).

NEXT UP

Next up, turning tendencies. The airplane is built stable, so what keeps pulling the nose to the left?

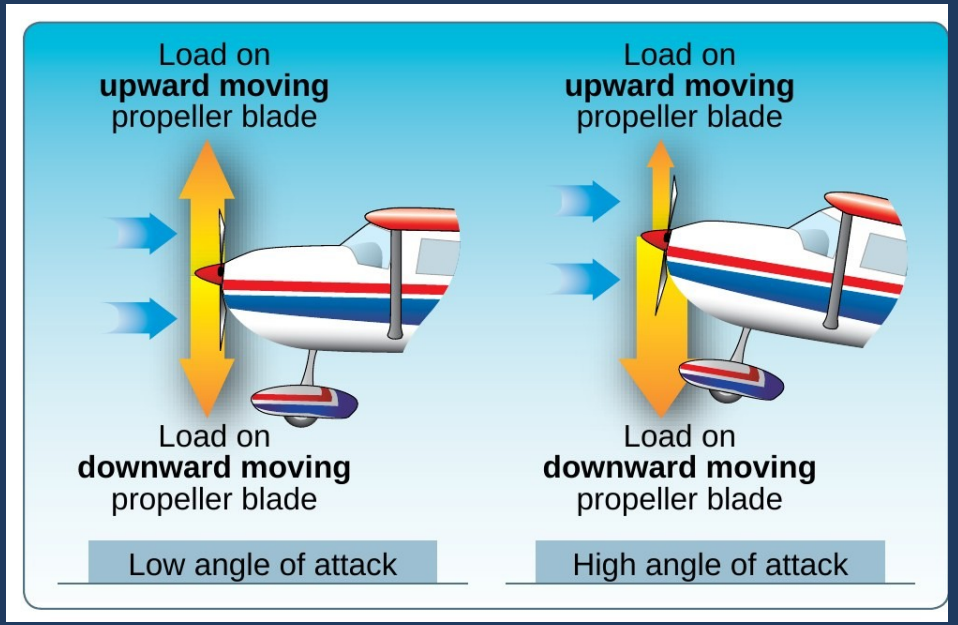
Source: PHAK 5-14 to 5-19

Step 5 of 6 | AI.II.D.K3

Turning Tendencies

What you will learn

- Torque reaction
- Corkscrewing slipstream
- Gyroscopic precession
- P-factor



Coming from step 4: the airplane is built to be stable, but the engine and propeller keep trying to pull it left.

Source: PHAK 5-30 to 5-32

Turning Tendencies: Torque Reaction

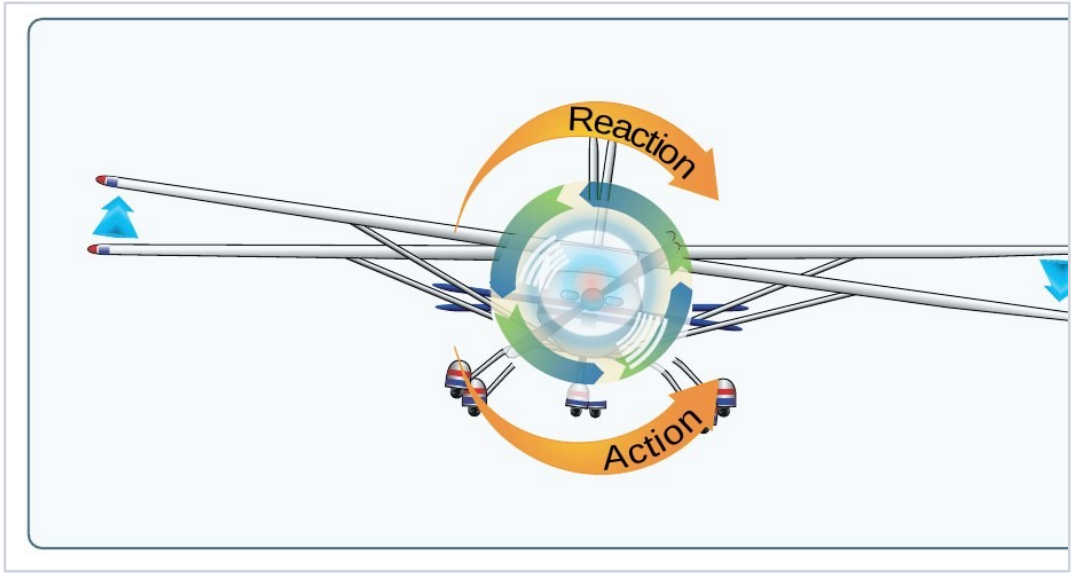
AI.II.D.K3

- **Four elements make up "torque"**
- **Prop turns clockwise (pilot view)**
- **Airplane rolls left in reaction**
- **On ground:** left yaw from tire drag

The left turning tendency called "torque" is made of four elements: torque reaction, corkscrewing slipstream, gyroscopic action, and P-factor (PHAK 5-30, PHAK 5-31).

Torque reaction uses Newton's Third Law: as the propeller turns one way, the airplane tries to roll the other way. Most U.S. engines turn the prop clockwise as seen from the pilot's seat (PHAK 5-31).

On the takeoff roll, more weight on the left main gear makes more friction and a left turning moment, corrected with rudder (PHAK 5-31).



KNOW THIS FOR THE CHECKRIDE

Four parts of "torque": torque reaction, slipstream, gyroscopic precession, P-factor.

Source: PHAK 5-30, PHAK 5-31, PHAK Fig 5-47

Turning Tendencies: Slipstream and Gyroscopic

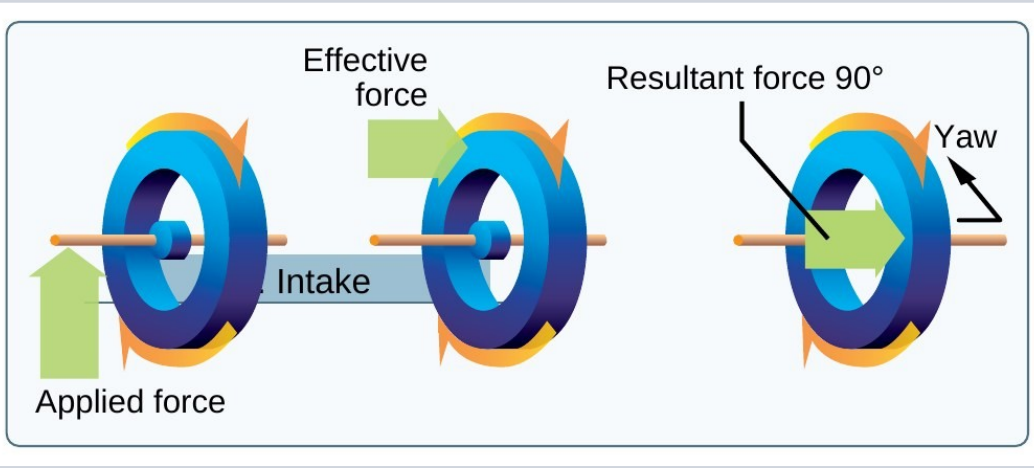
AI.II.D.K3

- **Slipstream hits left side of tail**
- **Strongest:** high power, low speed
- **Precession:** force acts 90 degrees ahead
- **Mostly tailwheel, when tail rises**

Corkscrew effect: the prop gives the slipstream a spiraling rotation. At high power and low speed it strikes the vertical fin and yaws the airplane (PHAK 5-31).

Gyroscopic precession: a force applied to a spinning propeller takes effect 90 degrees ahead in the direction of rotation (PHAK 5-31). Most prominent in tailwheel airplanes when the tail is raised on takeoff, causing a left yaw (PHAK 5-32).





Source: PHAK 5-31, PHAK 5-32, PHAK Fig 5-48, PHAK Fig 5-49

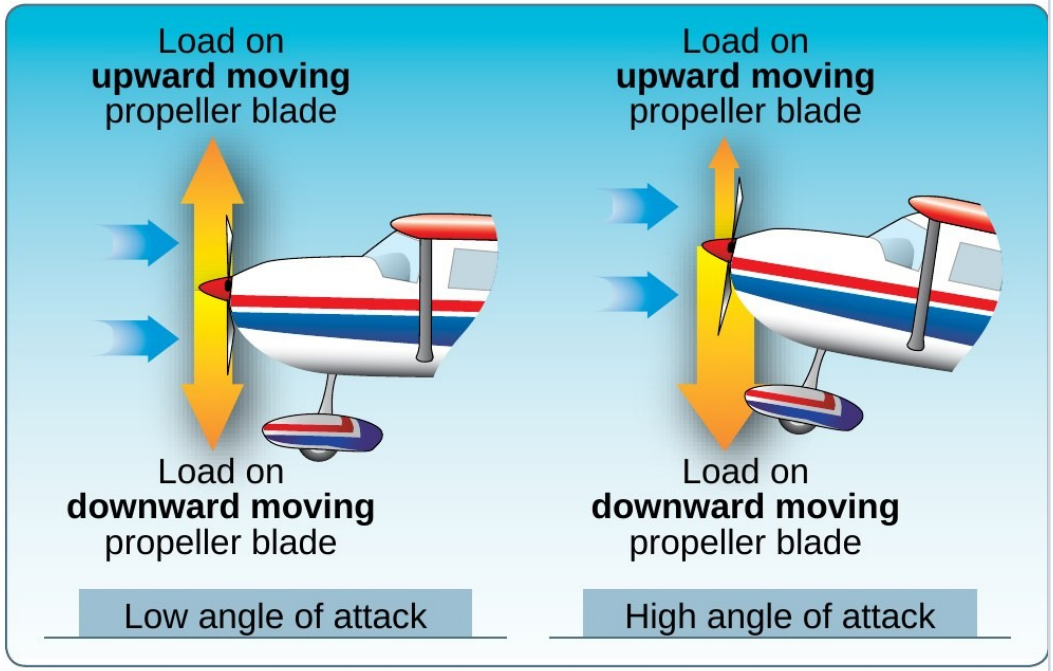
Turning Tendencies: P-Factor

AI.II.D.K3

- **High AOA (climb, slow flight)**
- **Down-going blade takes bigger bite**
- **More thrust on the right side**
- **Nose yaws left:** add right rudder

At a high AOA, the downward moving blade on the right has a greater "bite" than the upward moving blade. The center of thrust moves to the right of the prop disc, yawing the nose left (PHAK 5-32).

The four elements change with flight condition. The pilot must use the controls to compensate in every phase (PHAK 5-32).



EXAMPLE

C172S: full power climb at Vy. Feel the nose wanting to go left and hold right rudder to keep the ball centered.

Source: PHAK 5-32, PHAK Fig 5-51

Quick Check

AI.II.D.K3

1 **Name the four turning tendencies.**

Answer: Torque reaction, corkscrewing slipstream, gyroscopic precession, P-factor (PHAK 5-31).

2 **Gyroscopic precession: where does the resulting force act?**

Answer: 90 degrees ahead in the direction of rotation (PHAK 5-31).

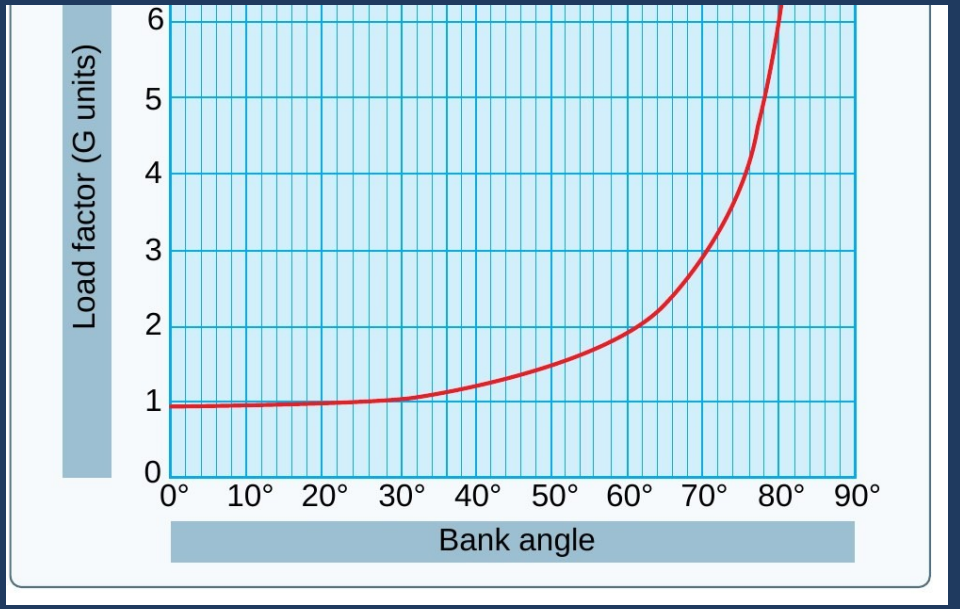
3 **In a full power climb, which way does the nose yaw and what fixes it?**

Answer: Left. Right rudder (PHAK 5-32).

NEXT UP

Next up, load factors. Now that we can control it, how hard can we maneuver before it stalls or breaks?

Load Factors



What you will learn

- Load factor and design limits
- Load factor in turns
- Load factor and stall speed
- Maneuvering speed

Coming from step 5: once we can control the airplane, how hard can we maneuver it before something breaks or stalls?

Source: PHAK 5-33 to 5-36, POH 2-4, POH 2-11

Definition: Load Factor

AI.II.D.K5

LOAD FACTOR

"The ratio of a specified load to the total weight of the aircraft."

IN PLAIN WORDS

Load factor is measured in Gs. 3 Gs means the structure carries three times the airplane's weight. C172S normal category: +3.8 to -1.52 Gs flaps up.

FLIGHT LOAD FACTOR LIMITS

NORMAL CATEGORY

Flight Load Factors (Maximum Takeoff Weight - 2550 POUNDS):
*Flaps Up . . . . . +3.8g, -1.52g
*Flaps Down . . . . . +3.0g

* The design load factors are 150% of the above, and in all cases, the structure meets or exceeds design loads.

UTILITY CATEGORY

Flight Load Factors (Maximum Takeoff Weight - 2200 POUNDS):
*Flaps Up . . . . . +4.4g, -1.76g
*Flaps Down . . . . . +3.0g

* The design load factors are 150% of the above, and in all cases, the structure meets or exceeds design loads.

Source: PHAK G-18, PHAK 5-33, PHAK 5-34, POH 2-11

Load Factor in Turns

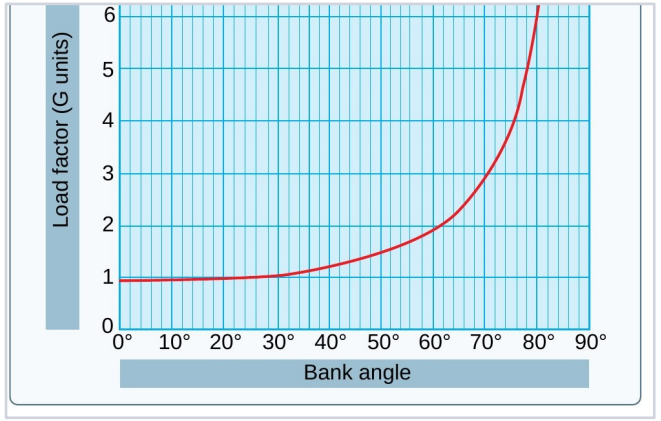
AI.II.D.K5

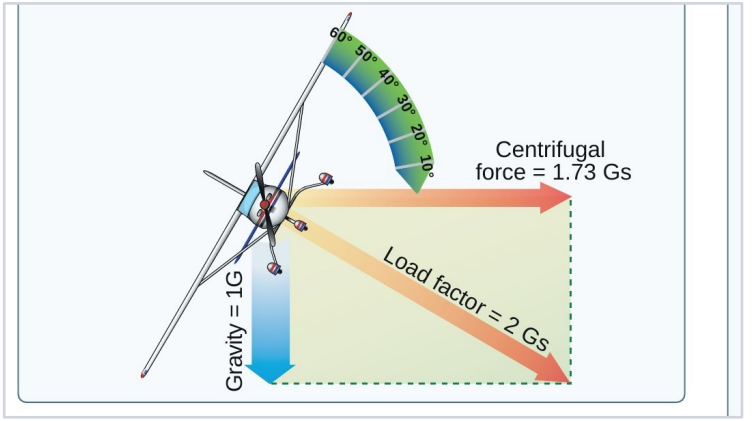
- **60 degree bank:** 2 Gs
- **Rises fast past 45 to 50 degrees**
- **About 70 to 75 degrees:** yield point
- **Wing must make that much lift**

In a coordinated, level turn, the load factor results from centrifugal force and weight (PHAK 5-34).

At 60 degrees of bank, load factor is 2 Gs. Load factor increases at a terrific rate after about 45 or 50 degrees (PHAK 5-34).

The yield factor of the average light plane is reached at about 70 to 75 degrees of bank (PHAK 5-36).





KNOW THIS FOR THE CHECKRIDE

60 degrees of bank = 2 Gs. Past 45 to 50 degrees it rises very fast (PHAK 5-34).

Source: PHAK 5-34, PHAK 5-36, PHAK Fig 5-53

Load Factor and Stall Speed

AI.II.D.K5

- **Stall speed rises with load**
- **Square root of load factor**
- **C172S:** 48 KIAS level
- **68 KIAS at 60 degrees bank**

Stalling speed increases in proportion to the square root of the load factor. A 50 knot stalling speed becomes 100 knots at 4 Gs (PHAK 5-35).

C172S, flaps up, 2,550 lb: 48 KIAS at 0 degrees, 52 at 30, 57 at 45, and 68 at 60 degrees of bank (POH Fig 5-3).

Altitude loss during stall recovery may be as much as 230 feet (POH Fig 5-3).

STALL SPEEDS AT 2550 POUNDS

CONDITIONS:
Power Off

MOST REARWARD CENTER OF GRAVITY

FLAP SETTING	ANGLE OF BANK							
	0°		30°		45°		60°	
	KIAS	KCAS	KIAS	KCAS	KIAS	KCAS	KIAS	KCAS
UP	48	53	52	57	57	63	68	75
10°	42	50	45	54	50	59	59	71
FULL	40	48	43	52	48	57	57	68

MOST FORWARD CENTER OF GRAVITY

FLAP SETTING	ANGLE OF BANK							
	0°		30°		45°		60°	
	KIAS	KCAS	KIAS	KCAS	KIAS	KCAS	KIAS	KCAS
UP	48	53	52	57	57	63	68	75
10°	43	51	46	55	51	61	61	72
FULL	40	48	43	52	48	57	57	68

NOTE

- Altitude loss during a stall recovery may be as much as 230 feet.
- KIAS values are approximate.

Source: PHAK 5-35, POH Fig 5-3

Maneuvering Speed (VA)

AI.II.D.K5

- **One full input, one axis**
- **C172S:** 105 KIAS at 2,550 lb
- **Lighter airplane:** lower VA
- **No protection from combined inputs**

VA is the speed below which you can move a single flight control, one time, to its full deflection, for one axis only, in smooth air, without risk of damage (PHAK 5-35).

Operating at or below VA does not provide structural protection against multiple full control inputs in one axis or full control inputs in more than one axis at the same time (PHAK 5-35).

C172S: 105 KIAS at 2,550 lb, 98 KIAS at 2,200 lb, 90 KIAS at 1,900 lb (POH 2-4).

Avoid rapid and large alternating control inputs; they may cause structural failure at any speed, including below VA (PHAK 5-36).

SYMBOL	SPEED	KCAS	KIAS	REMARKS
V_{NE}	Never Exceed Speed	160	163	Do not exceed this speed in any operation.
V_{NO}	Maximum Structural Cruising Speed	126	129	Do not exceed this speed except in smooth air, and then only with caution.
V_A	Maneuvering Speed: 2550 Pounds 2200 Pounds 1900 Pounds	 102 95 88	 105 98 90	Do not make full or abrupt control movements above this speed.
V_{FE}	Maximum Flap Extended Speed: Flaps 10° Flaps 10° to FULL	 107 85	 110 85	Do not exceed this speed with flaps down.
-----	Maximum Window Open Speed	160	163	Do not exceed this speed with windows open.

KNOW THIS FOR THE CHECKRIDE

VA is not protection against multiple full inputs or full inputs in more than one axis at once (PHAK 5-35).

Source: PHAK 5-35, PHAK 5-36, POH 2-4

Quick Check

AI.II.D.K5

1 **Load factor in a level 60 degree bank?**

Answer: 2 Gs (PHAK 5-34).

2 **C172S flaps up stall speed: wings level vs. 60 degrees of bank?**

Answer: 48 KIAS vs. 68 KIAS (POH Fig 5-3).

3 **C172S VA at 2,550 lb, and does it protect against two full inputs at once?**

Answer: 105 KIAS. No (POH 2-4, PHAK 5-35).

NEXT UP

Next up, risk management. Each principle comes with a risk. Let us manage them and plan how to teach this.

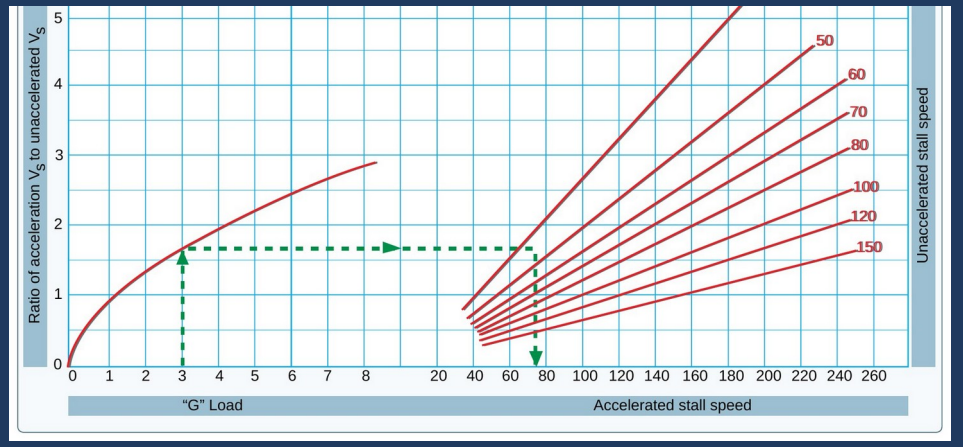
Source: PHAK 5-34, PHAK 5-35, POH 2-4, POH Fig 5-3

AI.II.D.R1, S1

Risk Management and Teaching It

What you will learn

- Stalls from load factor
- Wake, turning tendencies, spirals
- Teaching three elements in flight



Putting it together: every principle comes with a risk. Here is how to manage each one and how to teach it.

Source: PHAK 5-20, PHAK 5-35, AIM 7-4-6

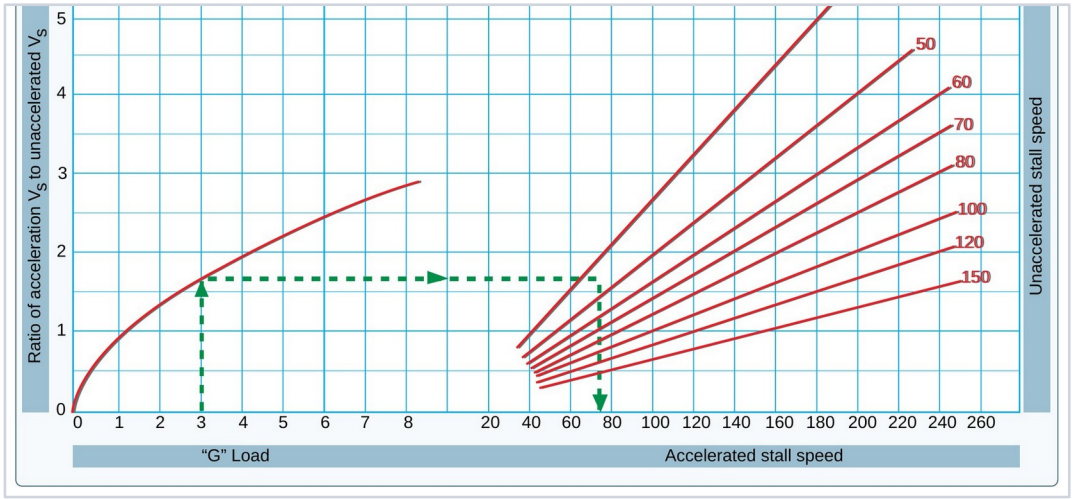
Risk: Stalls From Load Factor

AI.II.D.R1

- **Steep turns raise stall speed**
- **Base to final overshoot trap**
- **Keep bank shallow near ground**
- **Unload to recover**

Knowing load factors and VA helps prevent two of the most serious accident types: stalls from steep turns or excessive maneuvering near the ground, and structural failures from violent maneuvers or loss of control (PHAK 5-36).

Mitigation: plan the pattern to avoid overshoots, keep bank angles moderate near the ground, go around instead of tightening the turn, and reduce AOA at the first sign of a stall.



REAL WORLD SCENARIO

Your student overshoots the turn from base to final, steepens to 45 degrees, and pulls back. Ask what just happened to the stall speed and what they should do instead.

Source: PHAK 5-35, PHAK 5-36, POH Fig 5-3

Risk: Wake, Turning Tendencies, Spirals

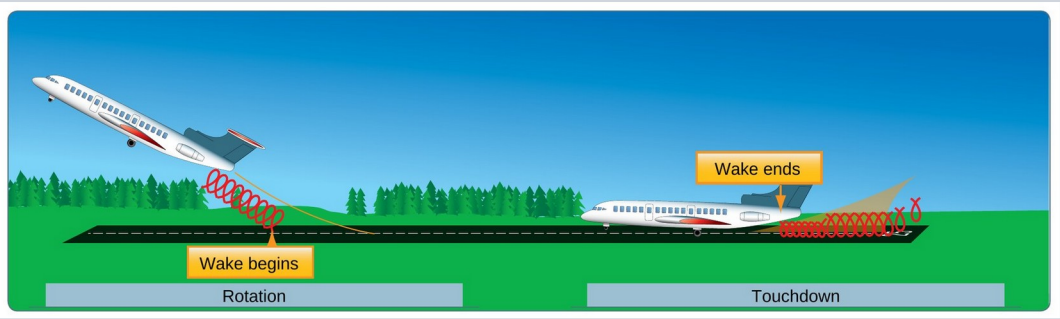
AI.II.D.R1

- **Wake:** plan your rotation point
- **Left yaw:** lead with right rudder
- **Spiral:** level wings, then pull
- **Know the airplane's numbers**

Wake turbulence: pick a rotation point before a larger aircraft's, and wait 2 minutes after a larger aircraft low approach or touch-and-go (AIM 7-4-6).

Turning tendencies: anticipate left yaw on takeoff and in climbs; use right rudder to stay on centerline and coordinated (PHAK 5-32).

Spiral instability: in a spiral, pulling back before leveling the wings only tightens the turn and raises load factor. Improper recovery from spirals leading to structural failure has contributed to many general aviation fatalities (PHAK 5-20).



Source: AIM 7-4-6, PHAK 5-20, PHAK 5-32

Teaching It: Three Elements in Flight

AI.II.D.S1

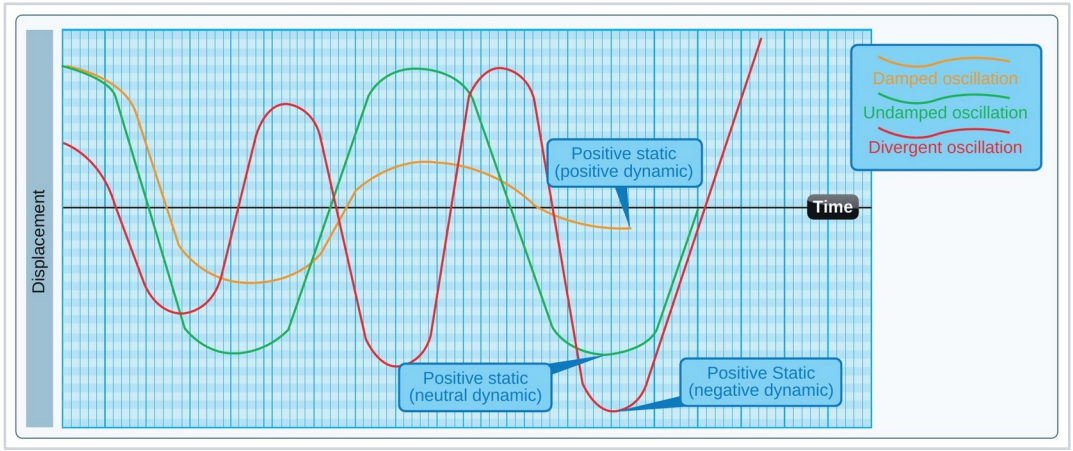
- **Stability:** trim, push, let go
- **P-factor:** climb, feel left yaw
- **Load factor:** 45 and 60 degree turns
- **Debrief each with the student**

S1 requires delivering instruction on at least three elements from K1 to K6. Example plan in the C172S at a safe altitude:

Stability (K2): trim for hands-off level flight, push the nose down slightly, release, and watch the oscillations damp out (PHAK 5-17).

Turning tendency (K3): in a full power climb, show the left yaw and the right rudder needed (PHAK 5-32).

Load factor (K5): fly 45 and 60 degree turns, feel the G load, and relate it to the higher stall speeds in the POH table (POH Fig 5-3).



Source: ACS AI.II.D.S1, PHAK 5-17, PHAK 5-32, POH Fig 5-3

Common Errors

AI.II.D

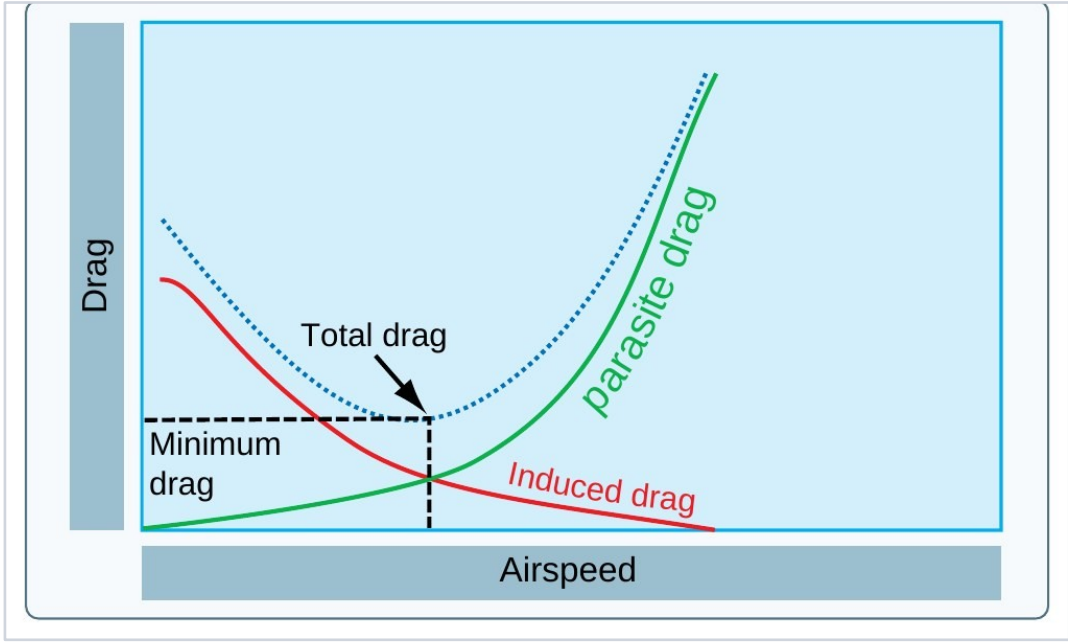
- **Saying all four forces are equal**
- **Stall "from low speed"**
- **Drag "inversely" (not squared)**
- **Treating VA as full protection**

"All four forces are equal": opposing forces cancel, but lift and weight are larger than thrust and drag (PHAK 5-2).

"Stalls happen at low speed": stalls happen when the critical AOA is exceeded, at any airspeed (PHAK G-9).

"Induced drag varies inversely with speed": it varies inversely with the square of the airspeed (PHAK 5-8).

"Below VA I am safe": VA protects only against one full input in one axis (PHAK 5-35).

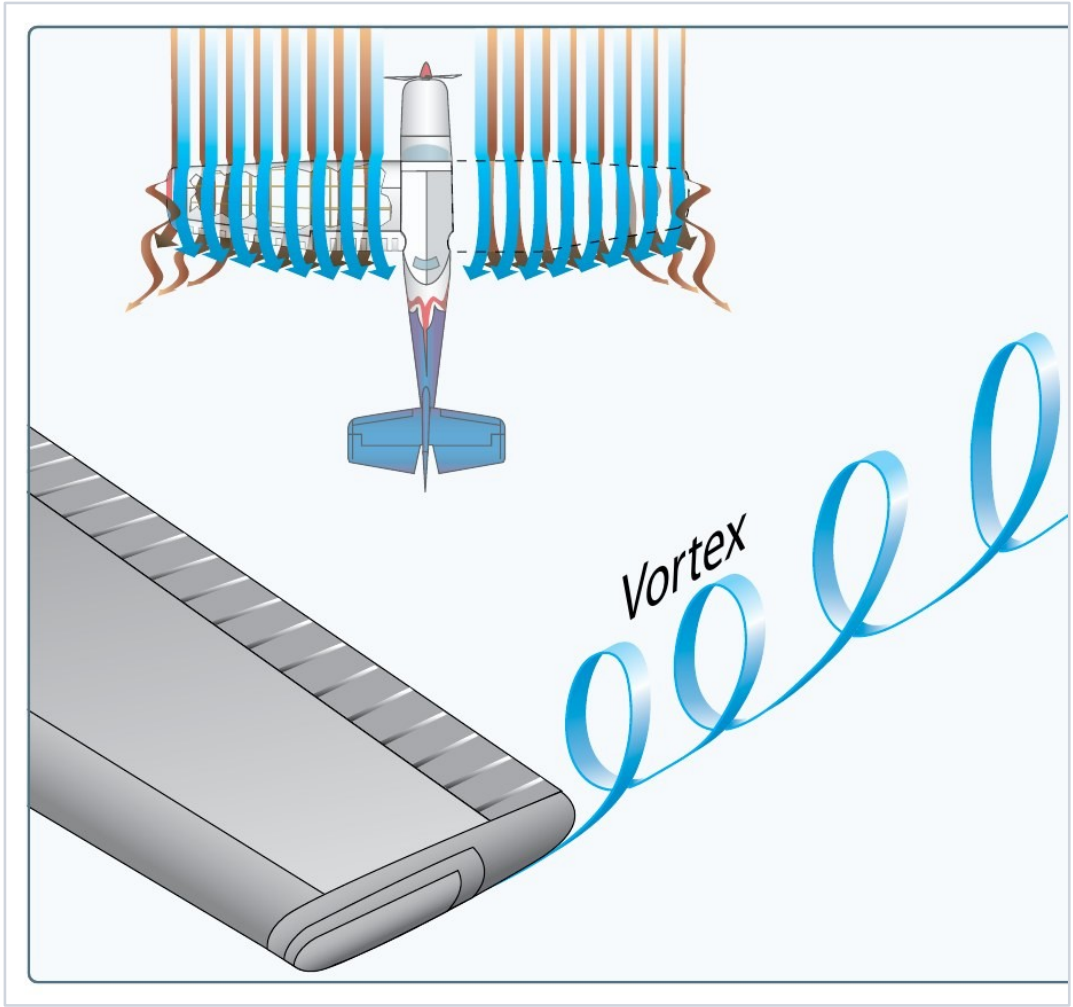


Source: PHAK 5-2, PHAK 5-8, PHAK G-9, PHAK 5-35

Review Questions

AI.II.D

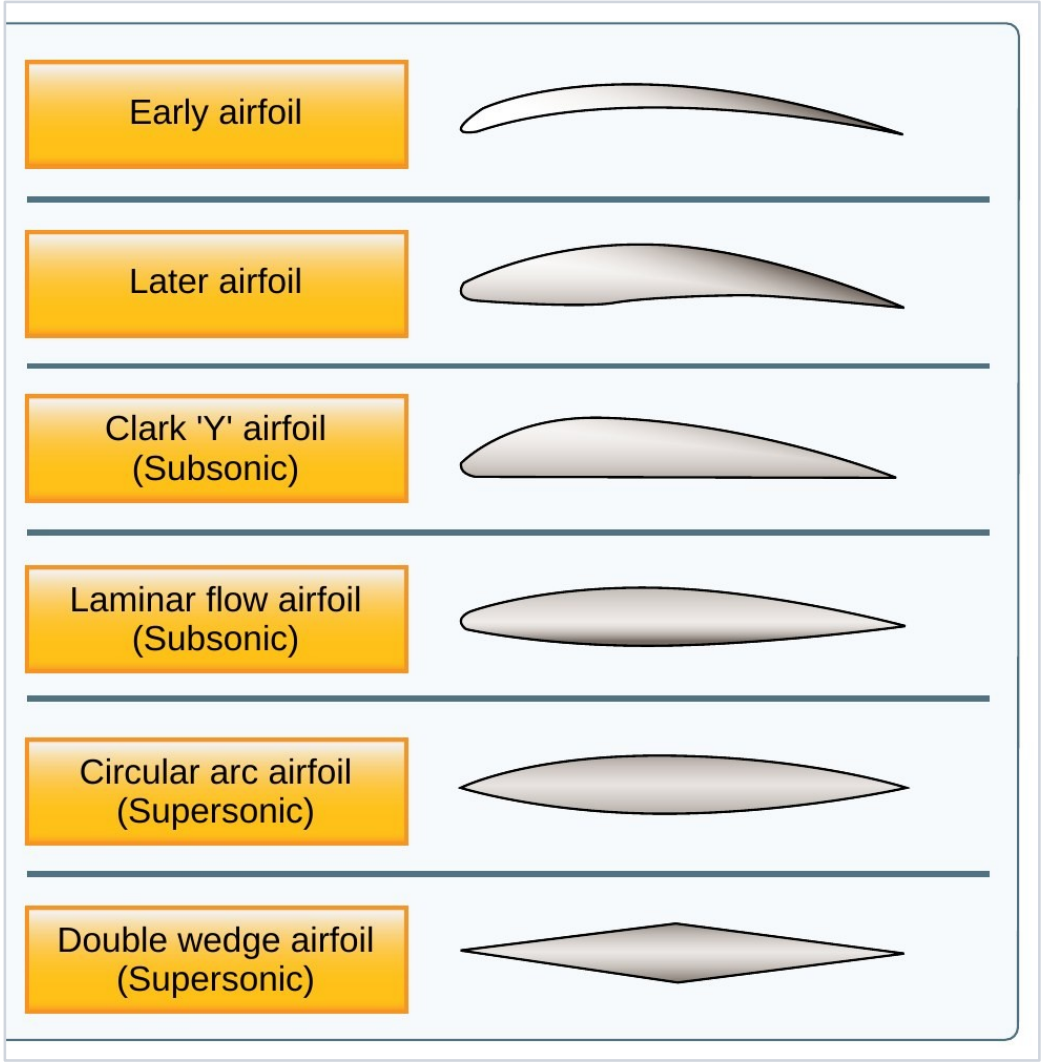
- **State Bernoulli's Principle**
- **What causes a stall?**
- **Load factor at 60 degrees?**
- **Departing behind a heavy:** where rotate?



Source: PHAK 4-6, PHAK G-9, PHAK 5-34, AIM 7-4-6

References

- PHAK Chapters 4 and 5
- AIM 7-4
- C172S POH Sections 2 and 5
- FAA-S-ACS-25



Source: See lesson plan for full list